

(19) **United States**

(12) **Patent Application Publication**

Ross et al.

(10) **Pub. No.: US 2025/0280457 A1**

(43) **Pub. Date:**

Sep. 4, 2025

(54) **SYSTEMS AND METHODS FOR IMPROVING WIRELESS MESH NETWORKS**

(71) Applicant: **L3VEL, LLC**, Wilmington, DE (US)

(72) Inventors: **Kevin Ross**, Lehi, UT (US);
Muhammad Ahsan Naim, Irving, TX (US)

(21) Appl. No.: **19/031,133**

(22) Filed: **Jan. 17, 2025**

H04W 24/02 (2009.01)
H04W 28/02 (2009.01)
H04W 40/04 (2009.01)
H04W 40/20 (2009.01)
H04W 84/02 (2009.01)
H04W 84/22 (2009.01)
H04W 88/10 (2009.01)
H04W 88/12 (2009.01)
H04W 88/14 (2009.01)
H04W 92/02 (2009.01)
H04W 92/04 (2009.01)
H04W 92/16 (2009.01)

Related U.S. Application Data

(63) Continuation of application No. 18/479,145, filed on Oct. 2, 2023, now Pat. No. 12,267,901, which is a continuation of application No. 17/665,790, filed on Feb. 7, 2022, now Pat. No. 11,778,678, which is a continuation of application No. 17/345,914, filed on Jun. 11, 2021, now Pat. No. 11,246,180, which is a continuation of application No. 16/680,457, filed on Nov. 11, 2019, now Pat. No. 11,044,617, which is a continuation-in-part of application No. 16/590,217, filed on Oct. 1, 2019, now Pat. No. 11,102,834, which is a continuation-in-part of application No. 16/508,289, filed on Jul. 10, 2019, now Pat. No. 10,966,266.

(60) Provisional application No. 62/856,697, filed on Jun. 3, 2019, provisional application No. 62/833,485, filed on Apr. 12, 2019, provisional application No. 62/771,508, filed on Nov. 26, 2018, provisional application No. 62/753,885, filed on Oct. 31, 2018, provisional application No. 62/696,688, filed on Jul. 11, 2018.

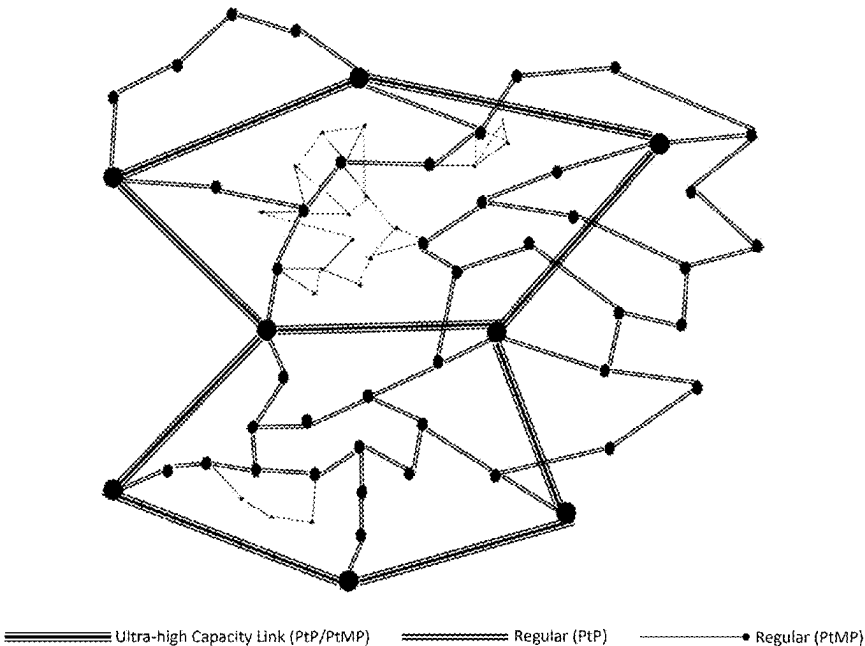
Publication Classification

(51) **Int. Cl.**
H04W 76/15 (2018.01)
H04W 16/28 (2009.01)

(52) **U.S. Cl.**
CPC *H04W 76/15* (2018.02); *H04W 16/28* (2013.01); *H04W 24/02* (2013.01); *H04W 28/021* (2013.01); *H04W 40/04* (2013.01); *H04W 40/20* (2013.01); *H04W 84/02* (2013.01); *H04W 84/22* (2013.01); *H04W 88/10* (2013.01); *H04W 88/12* (2013.01); *H04W 88/14* (2013.01); *H04W 92/02* (2013.01); *H04W 92/04* (2013.01); *H04W 92/16* (2013.01)

(57) **ABSTRACT**

Disclosed herein is a wireless mesh network comprised of ultra-high-capacity nodes that are capable of establishing ultra-high-capacity links (e.g., point-to-point or point-to-multipoint bi-directional communication links) using a millimeter wave spectrum, including but not limited to 28 Ghz, 39 Ghz, 37/42 Ghz, 60 Ghz (including V band), or E-band frequencies, as examples. The higher capacity and/or extended range of these ultra-high-capacity nodes/links may be achieved via various advanced signal processing techniques. Further, these ultra-high-capacity nodes/links may be used in conjunction with other types of point-to-point and/or point-to-multipoint links to build a multi-layer wireless mesh network.



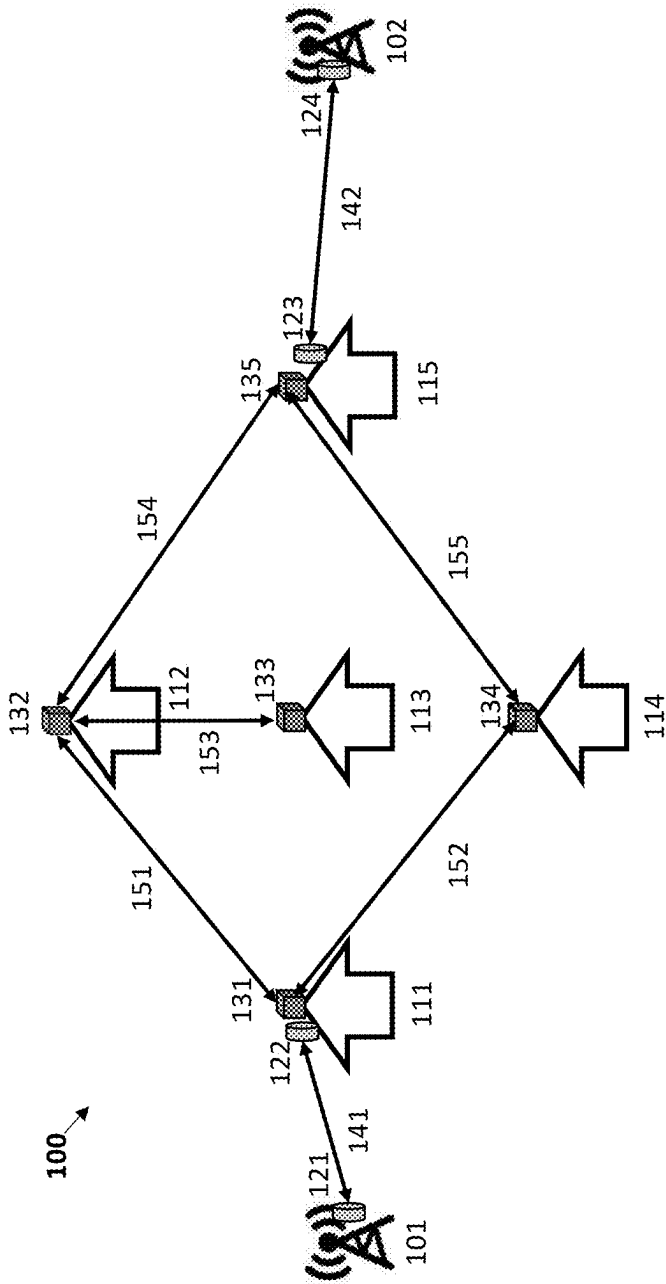


FIG. 1

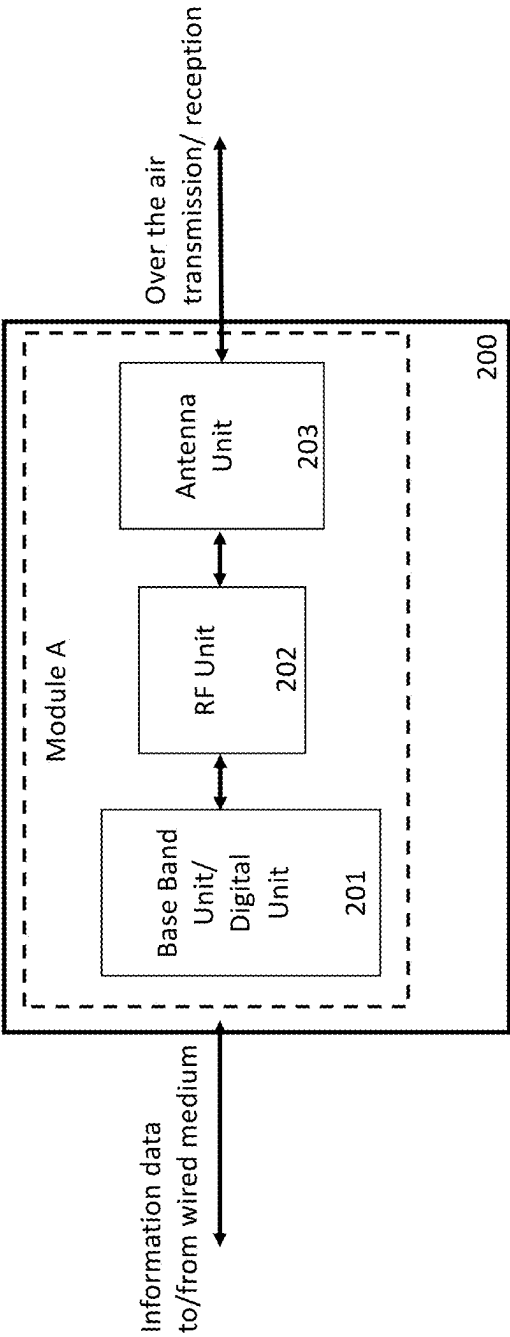


FIG. 2

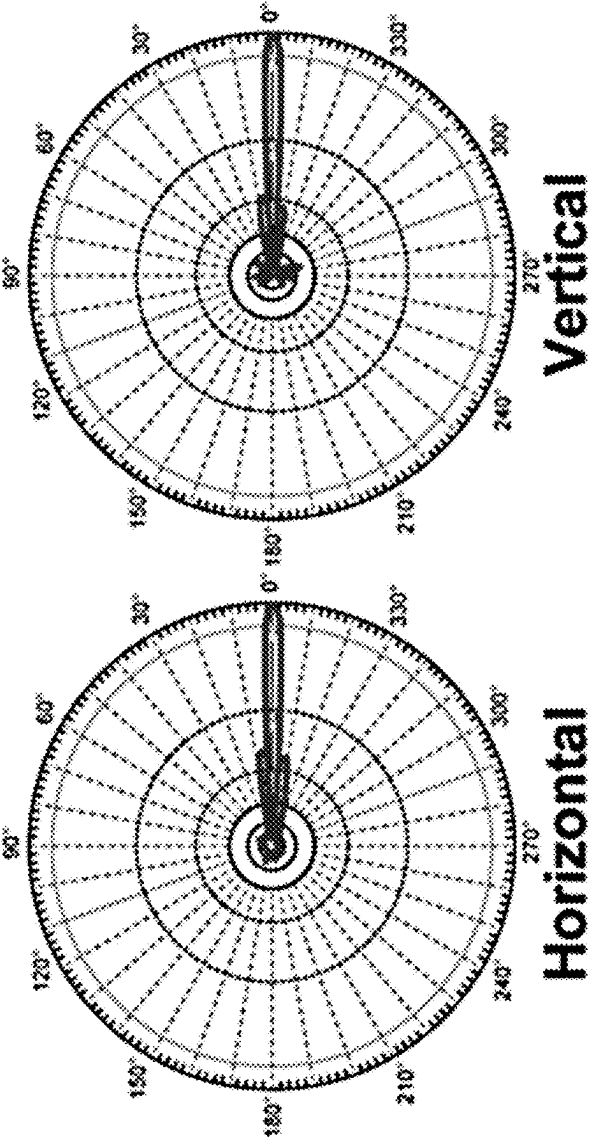


FIG. 3

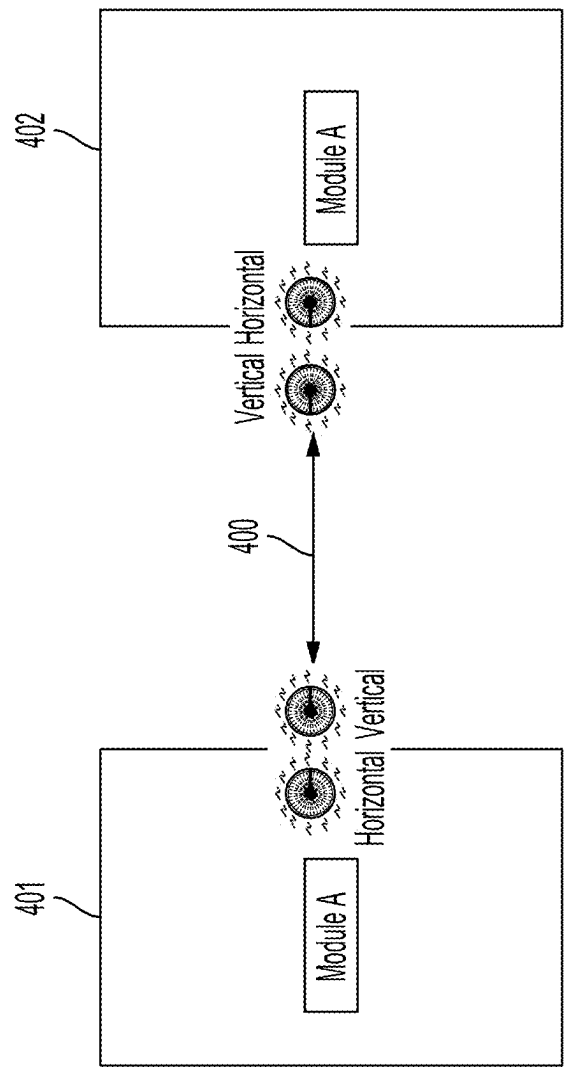


FIG. 4

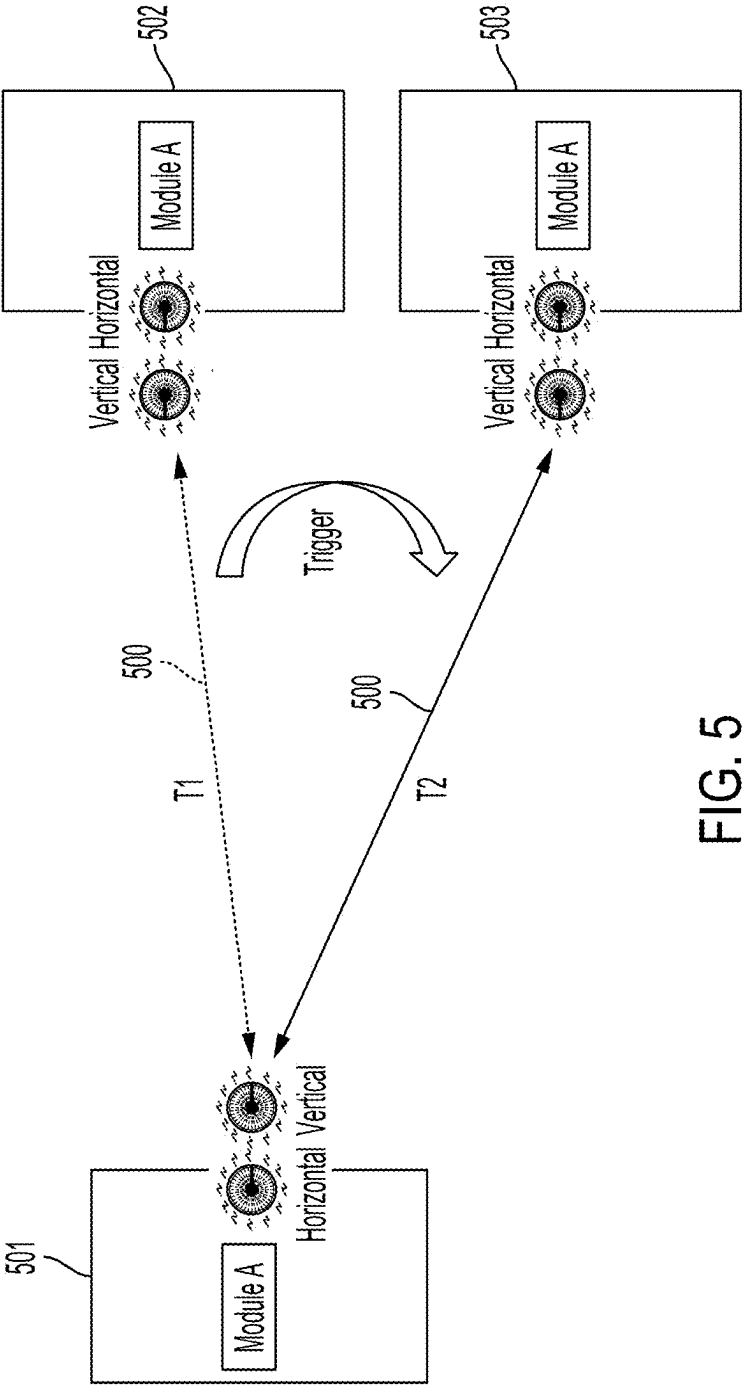


FIG. 5

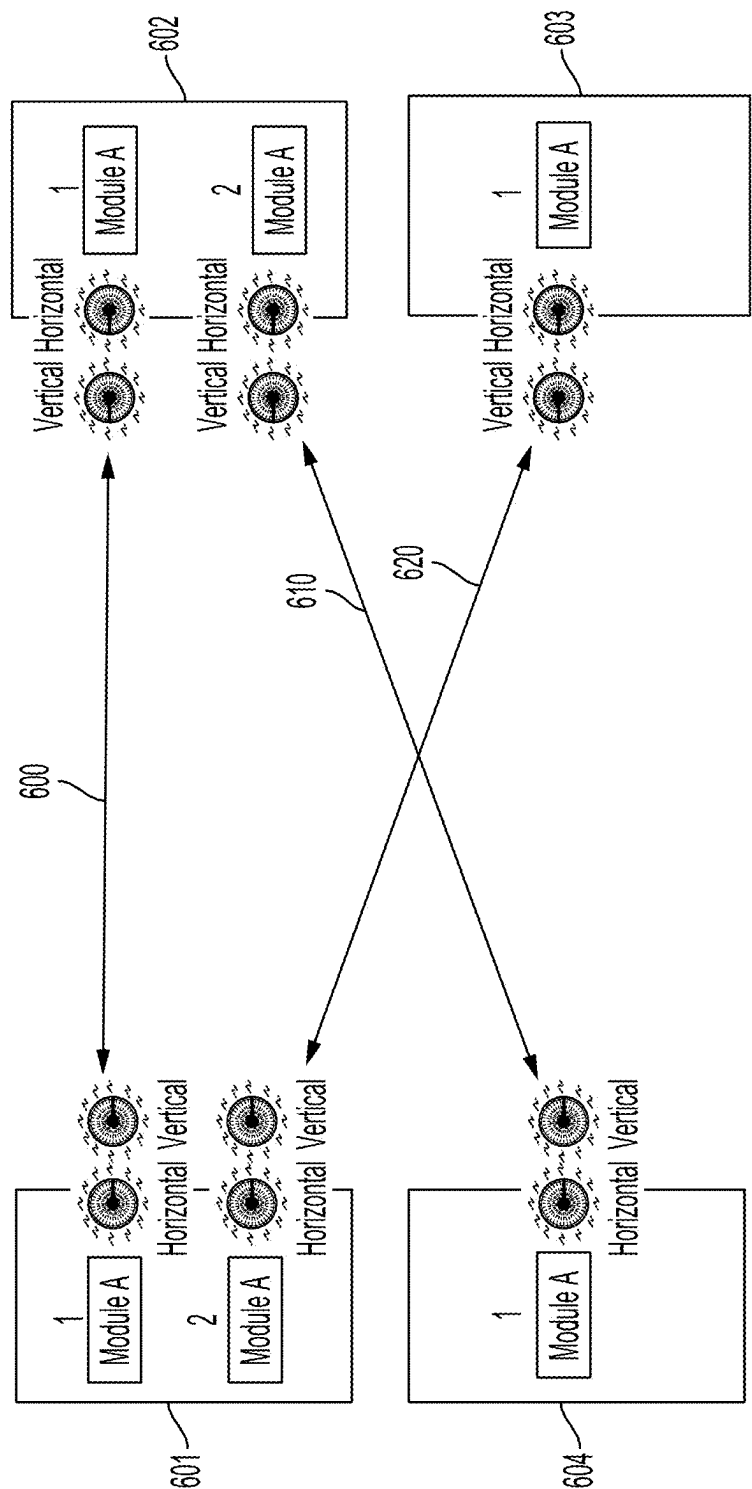


FIG. 6

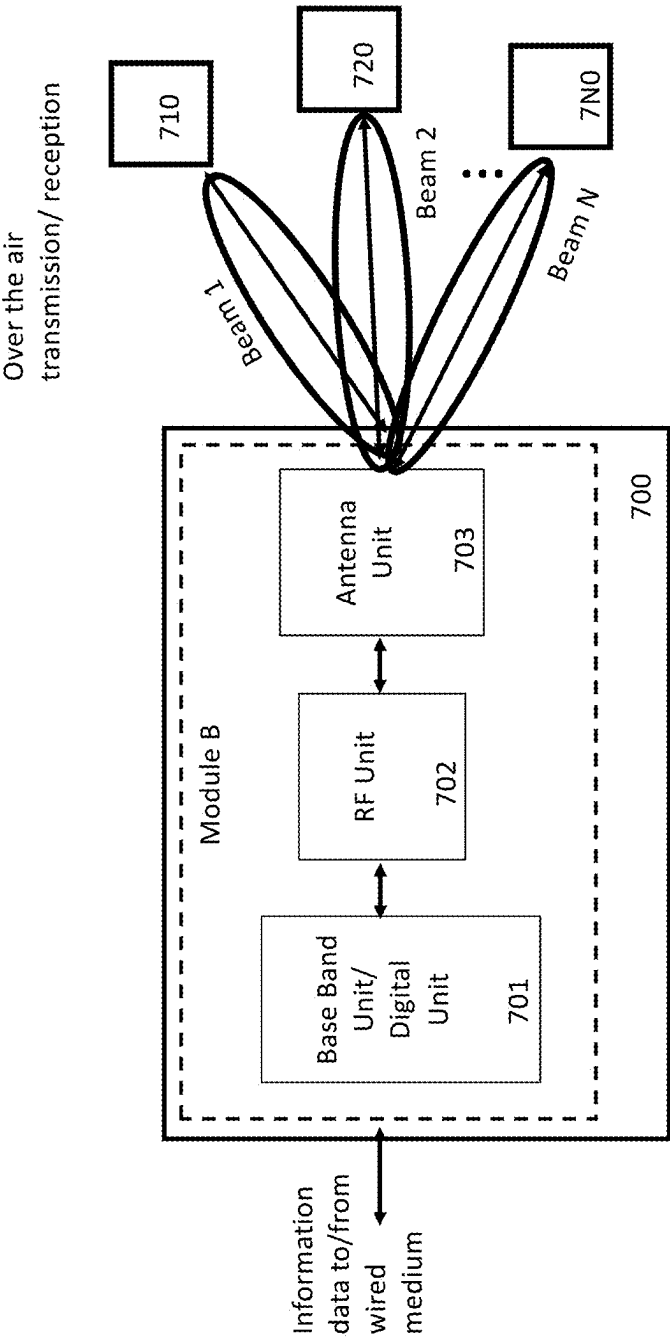


FIG. 7

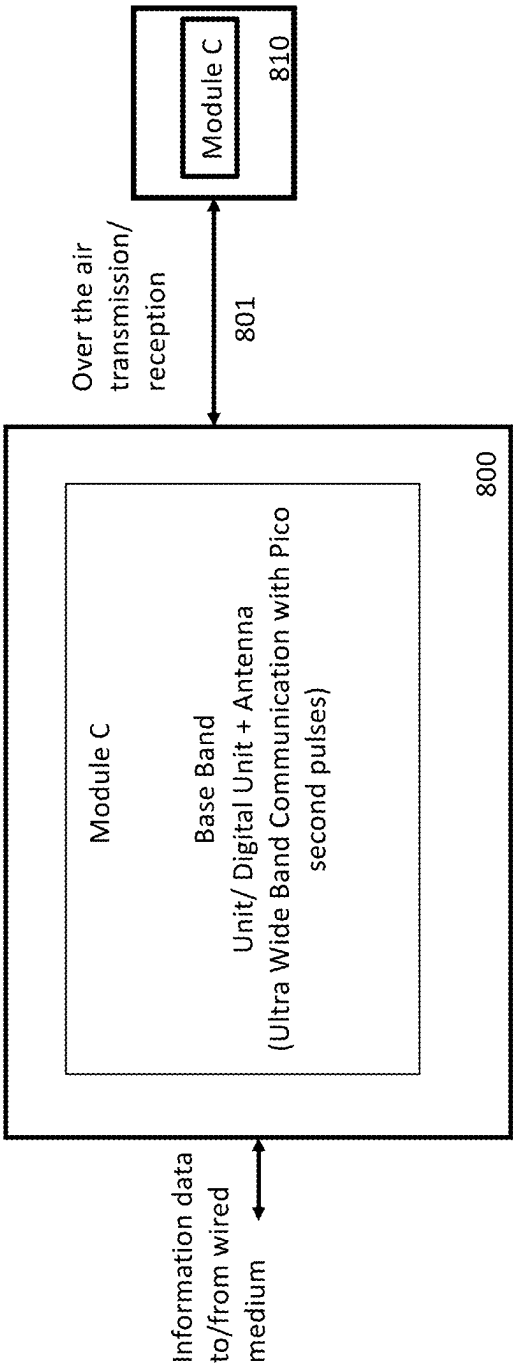


FIG. 8

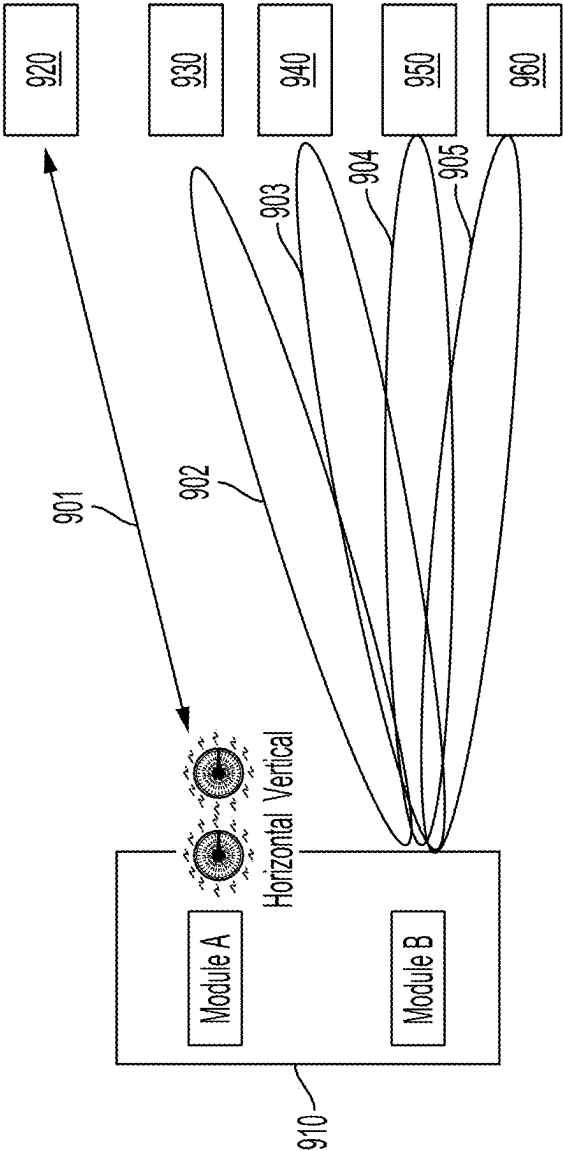


FIG. 9

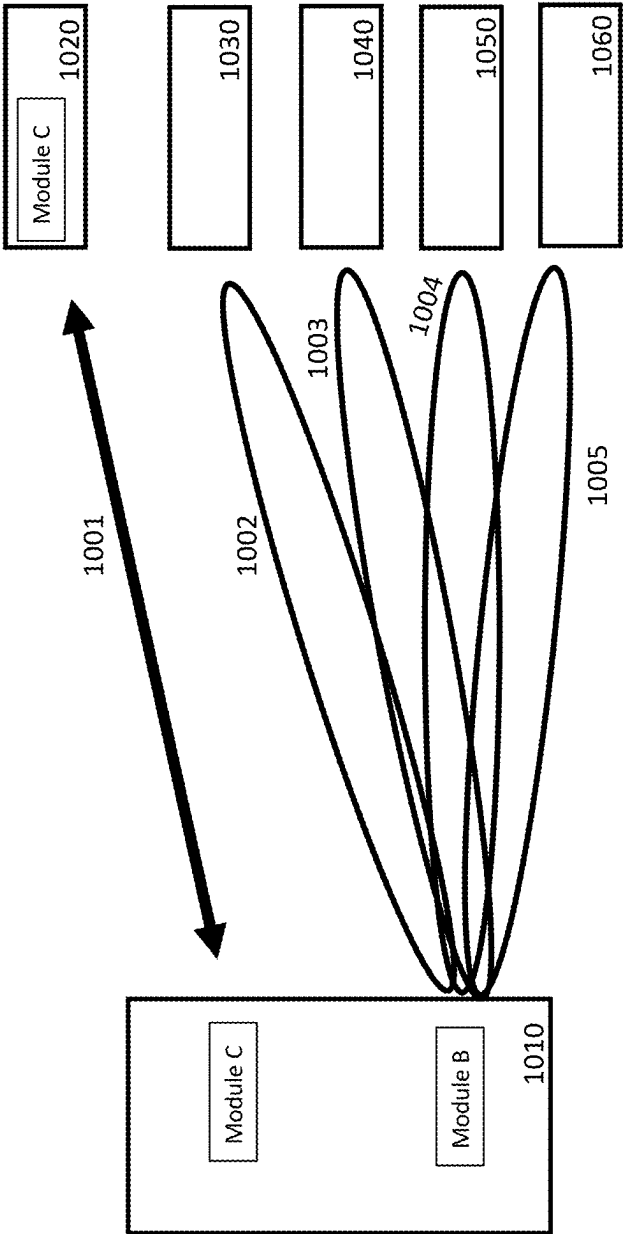


FIG. 10

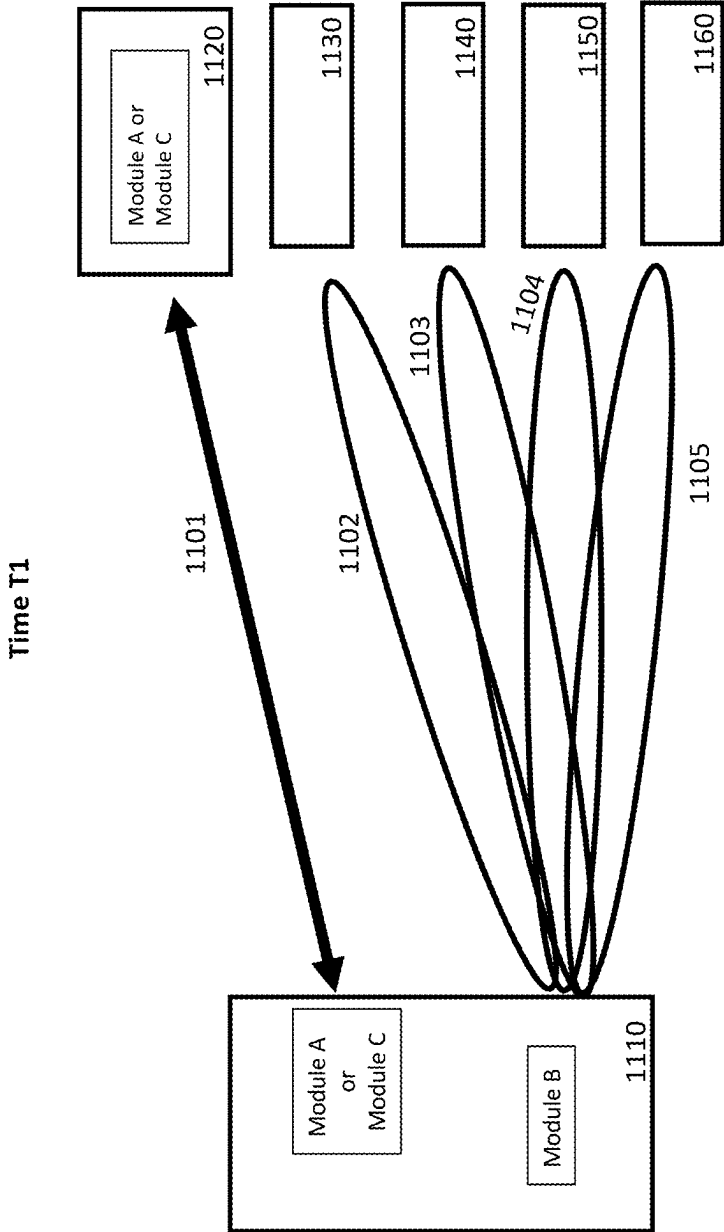


FIG. 11A

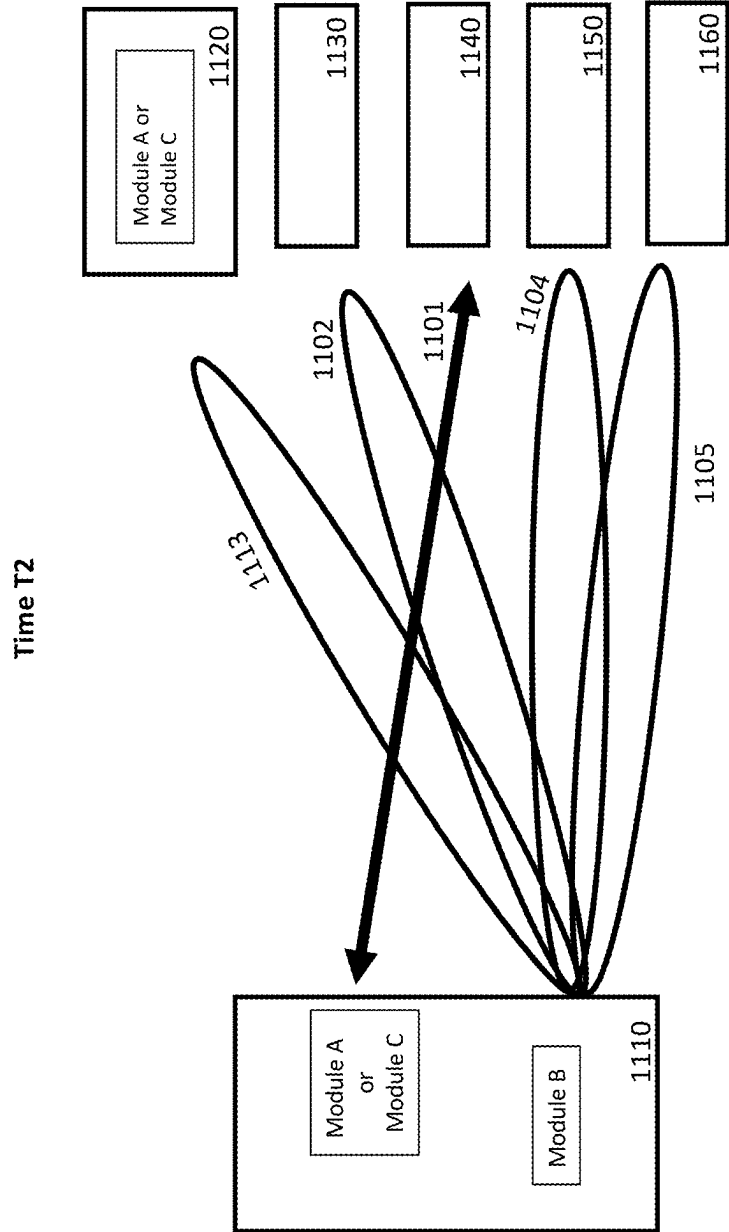


FIG. 11B

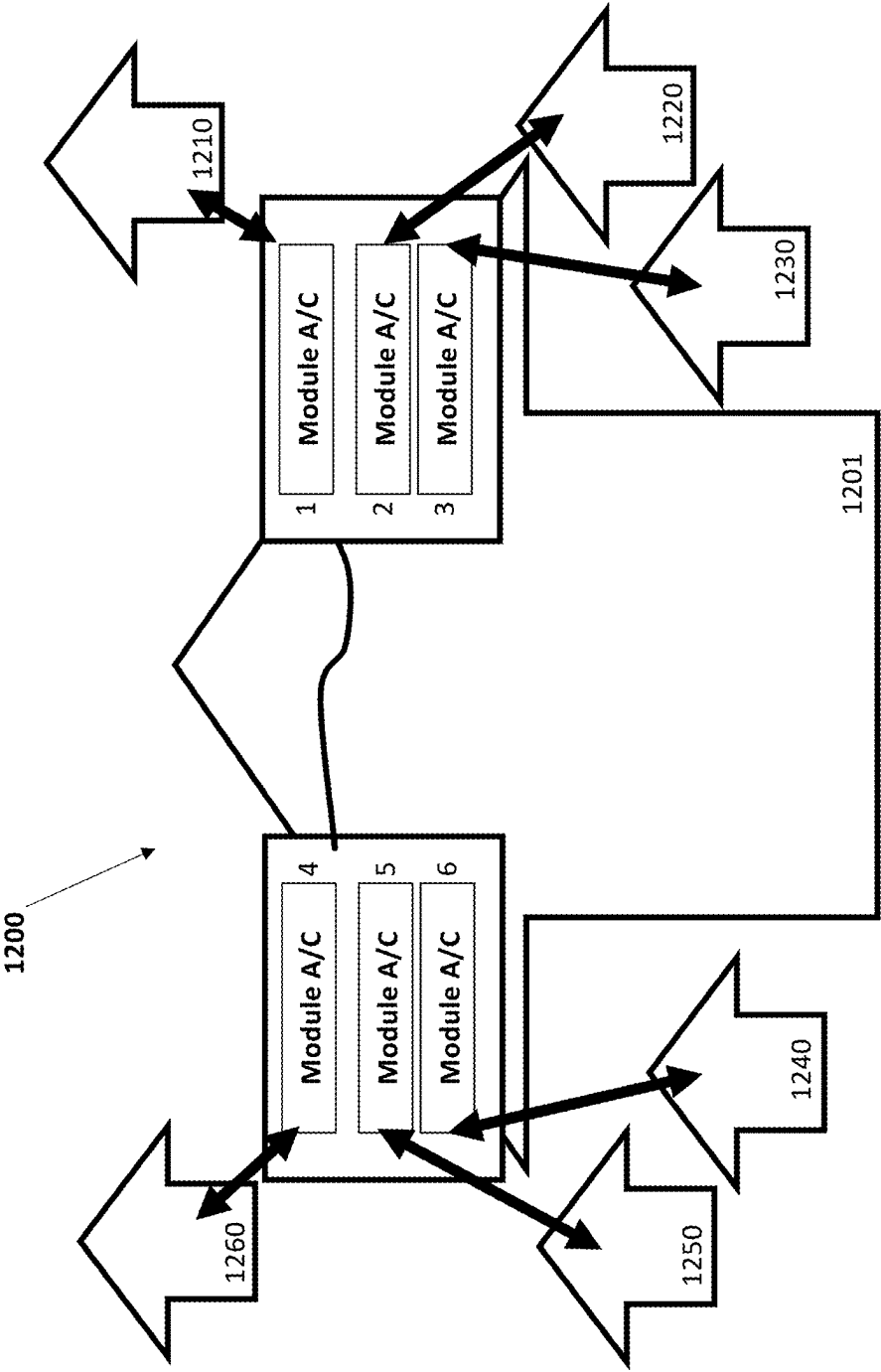


FIG. 12

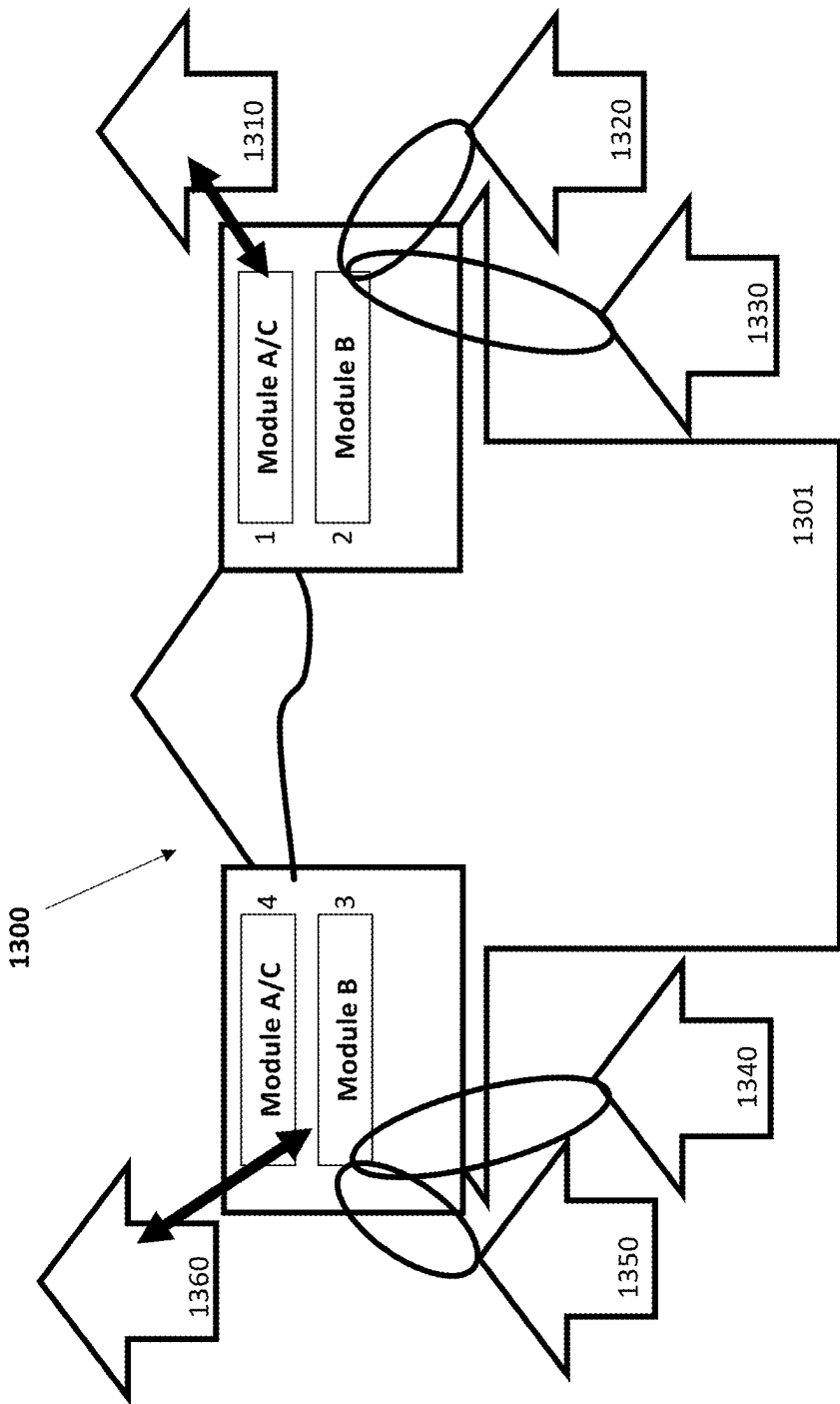


FIG. 13

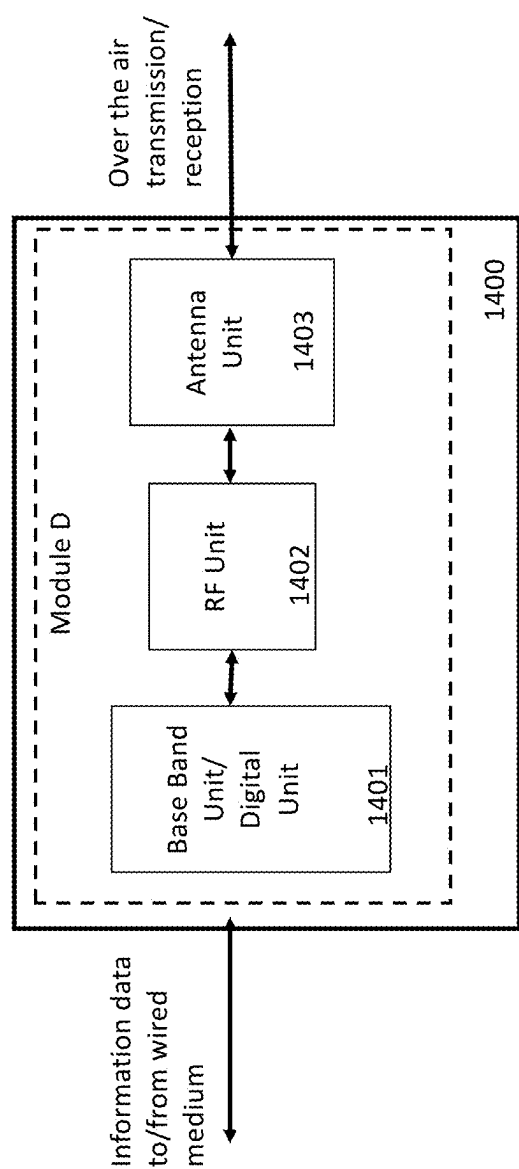


FIG. 14

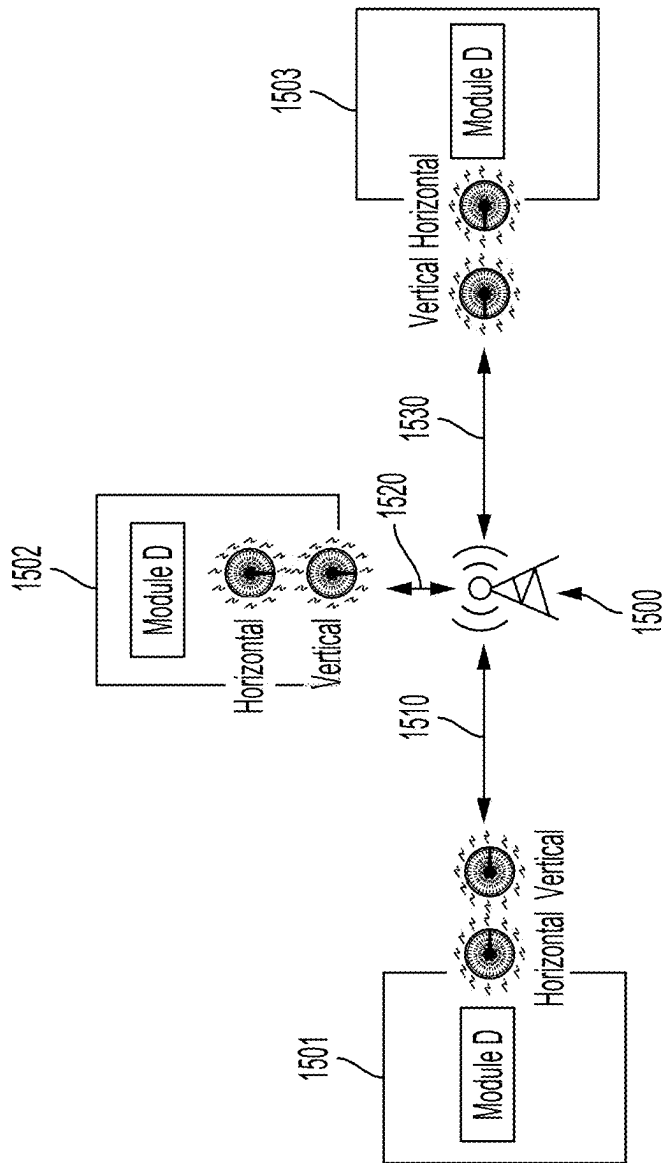


FIG. 15

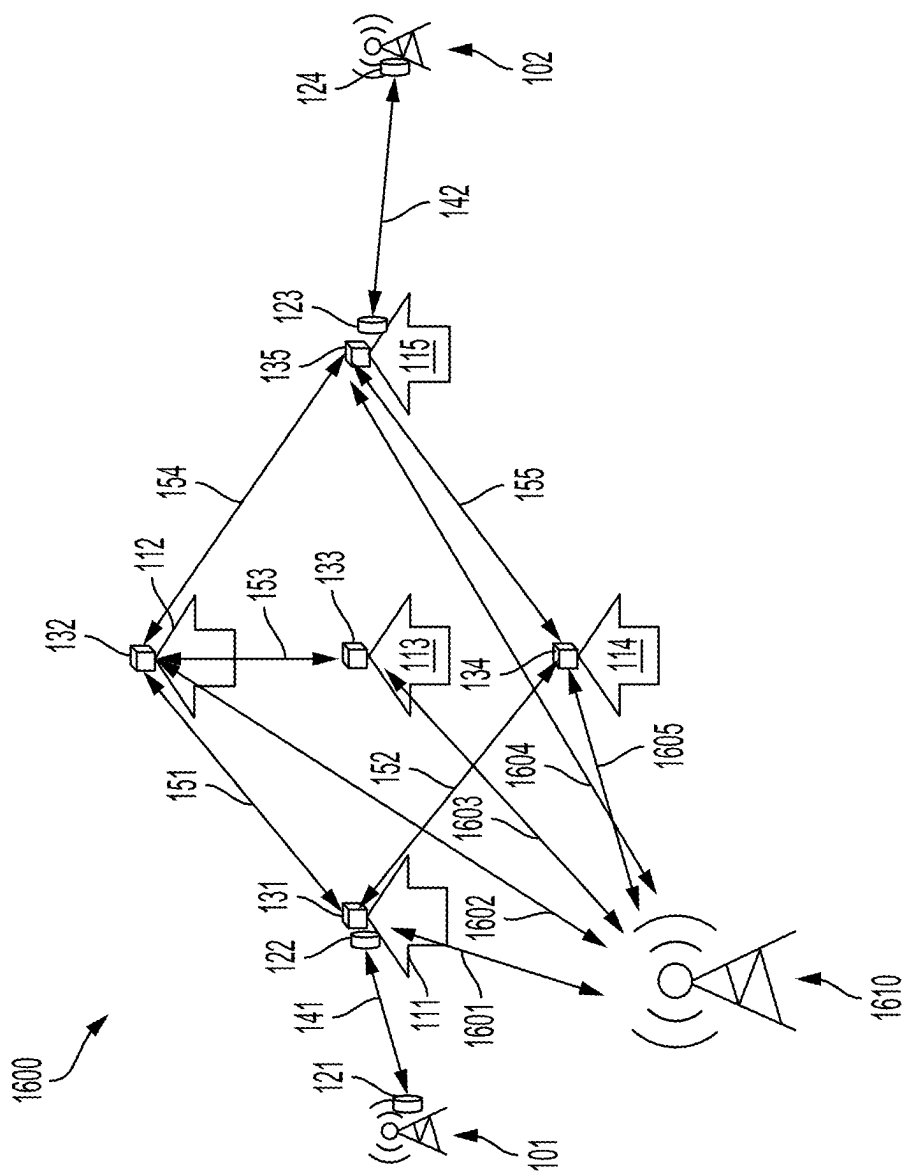


FIG. 16

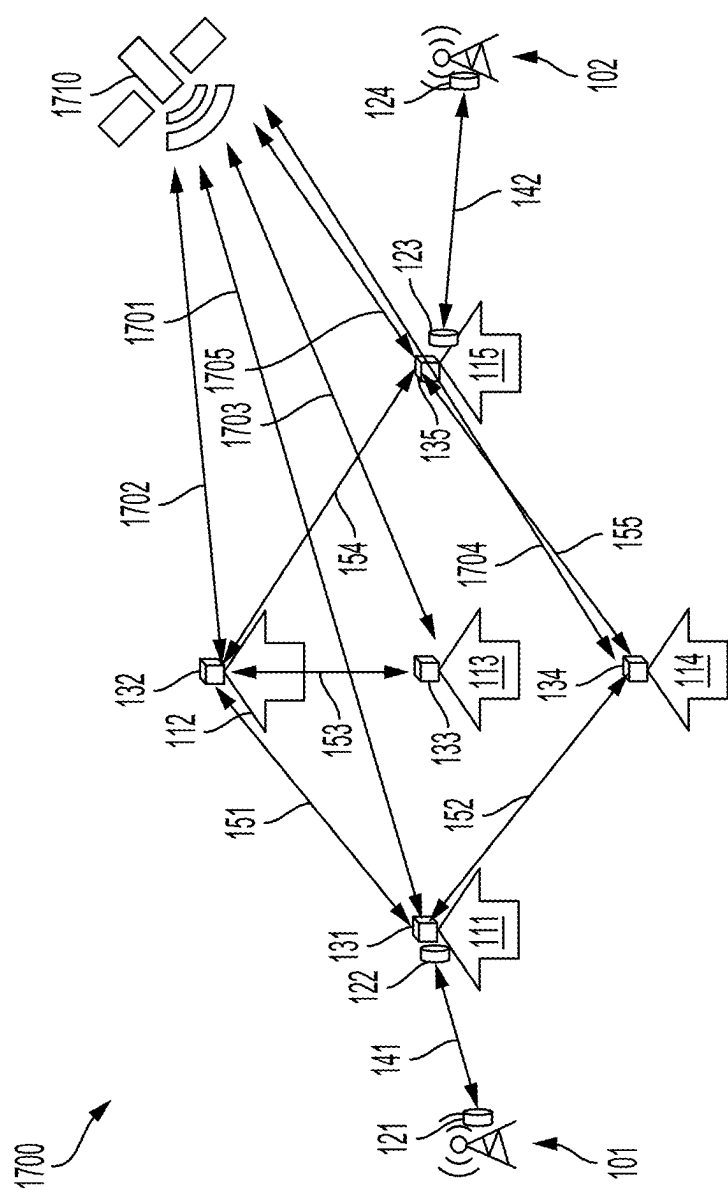


FIG. 17

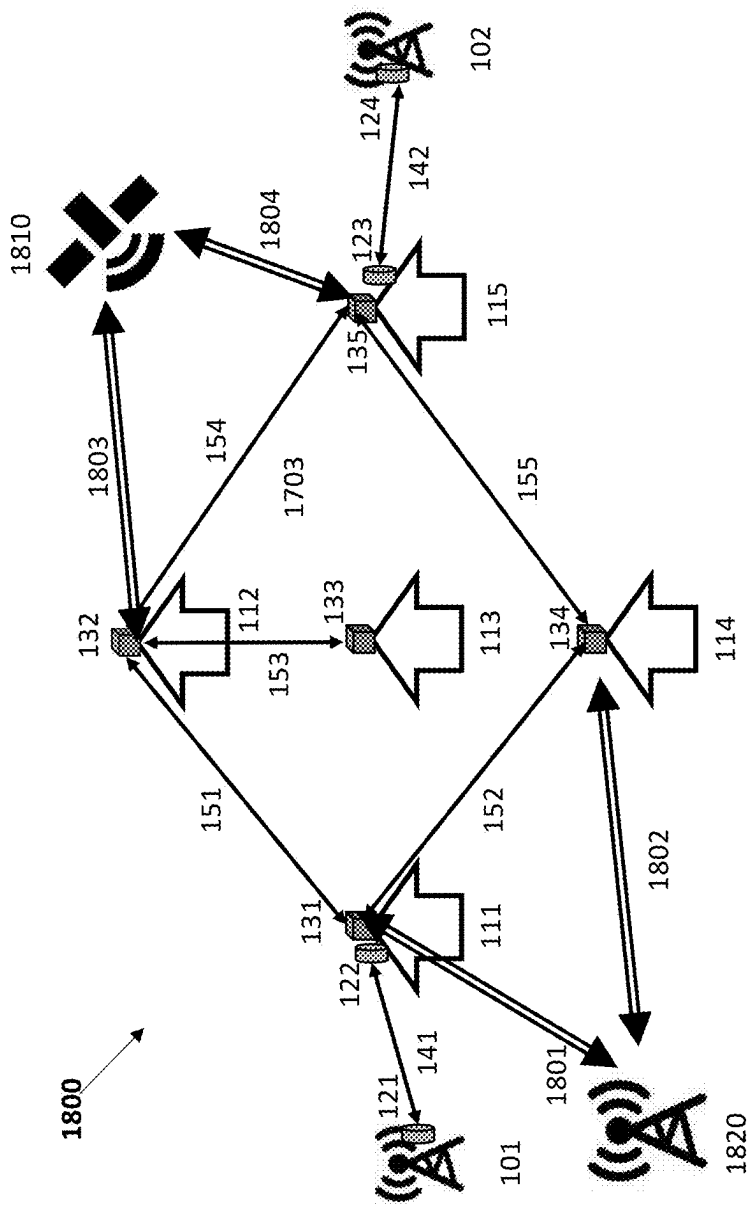


FIG. 18

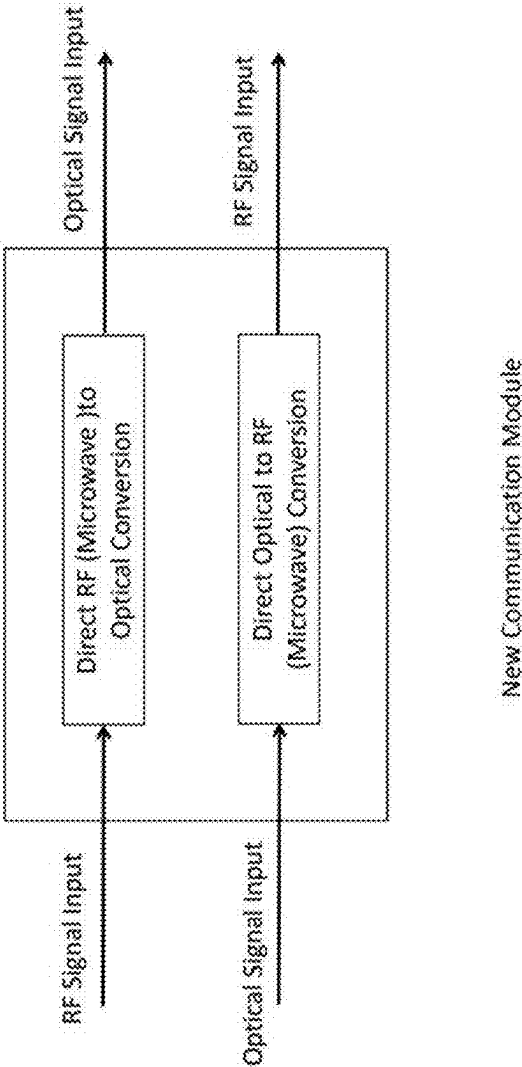


FIG. 19

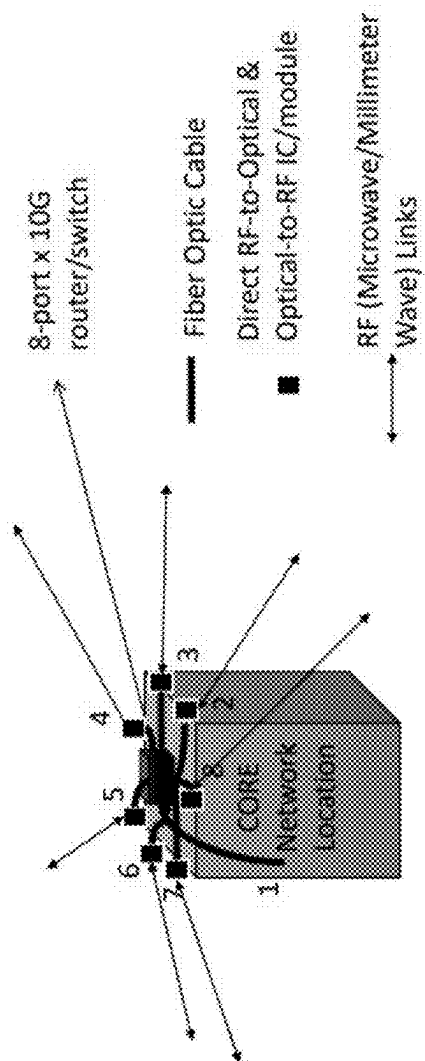


FIG. 20

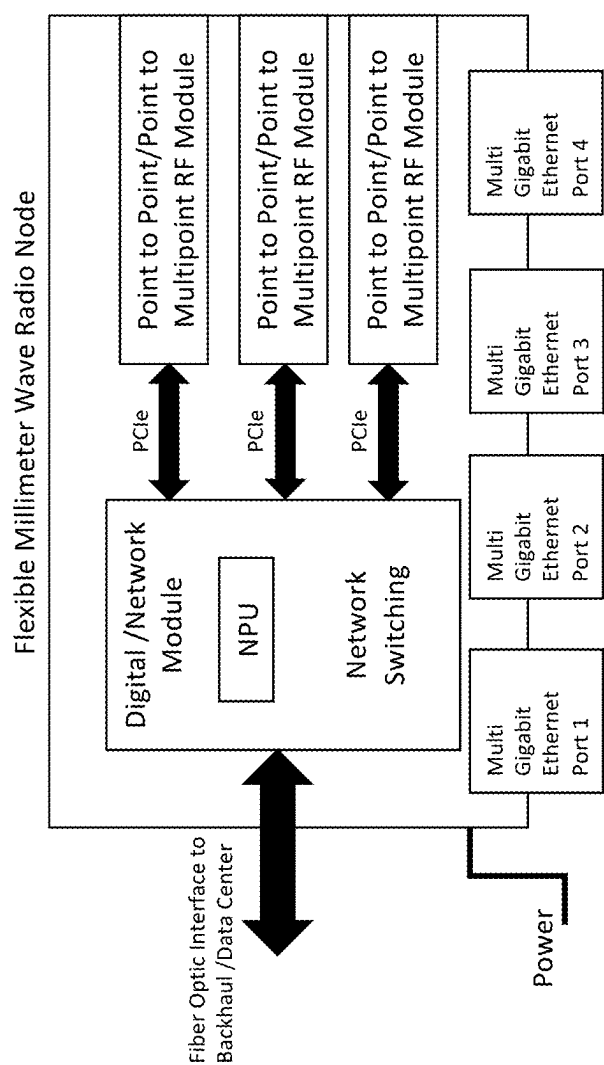


FIG. 21

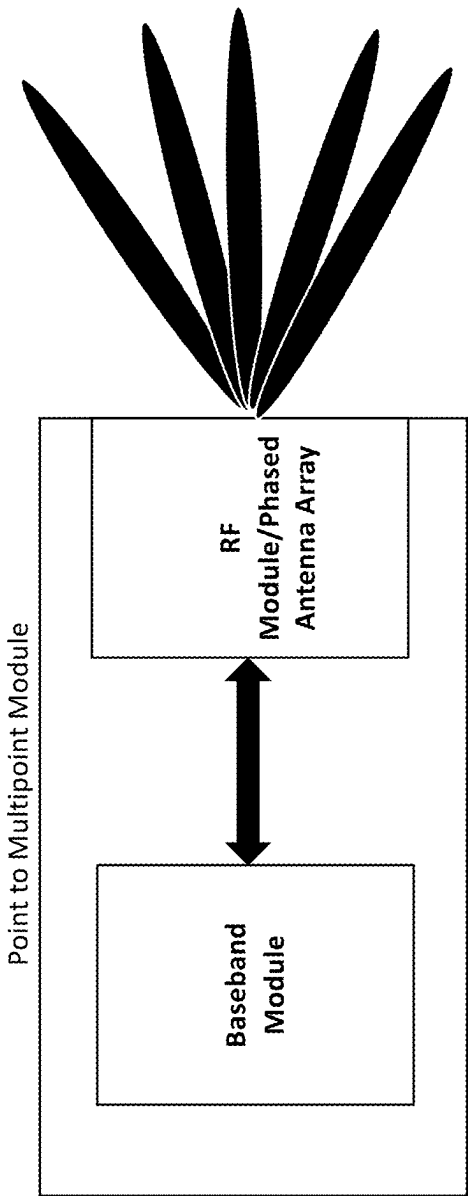


FIG. 22

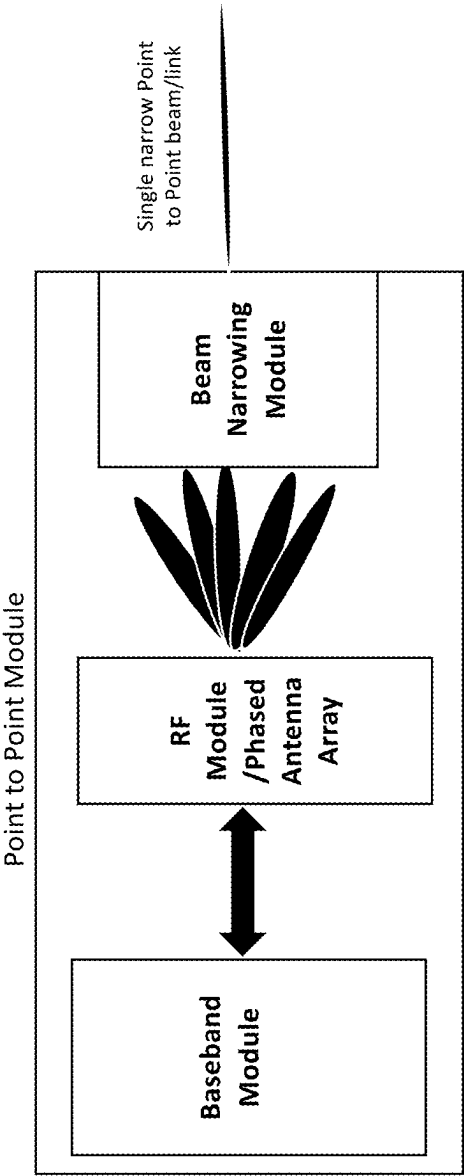


FIG. 23

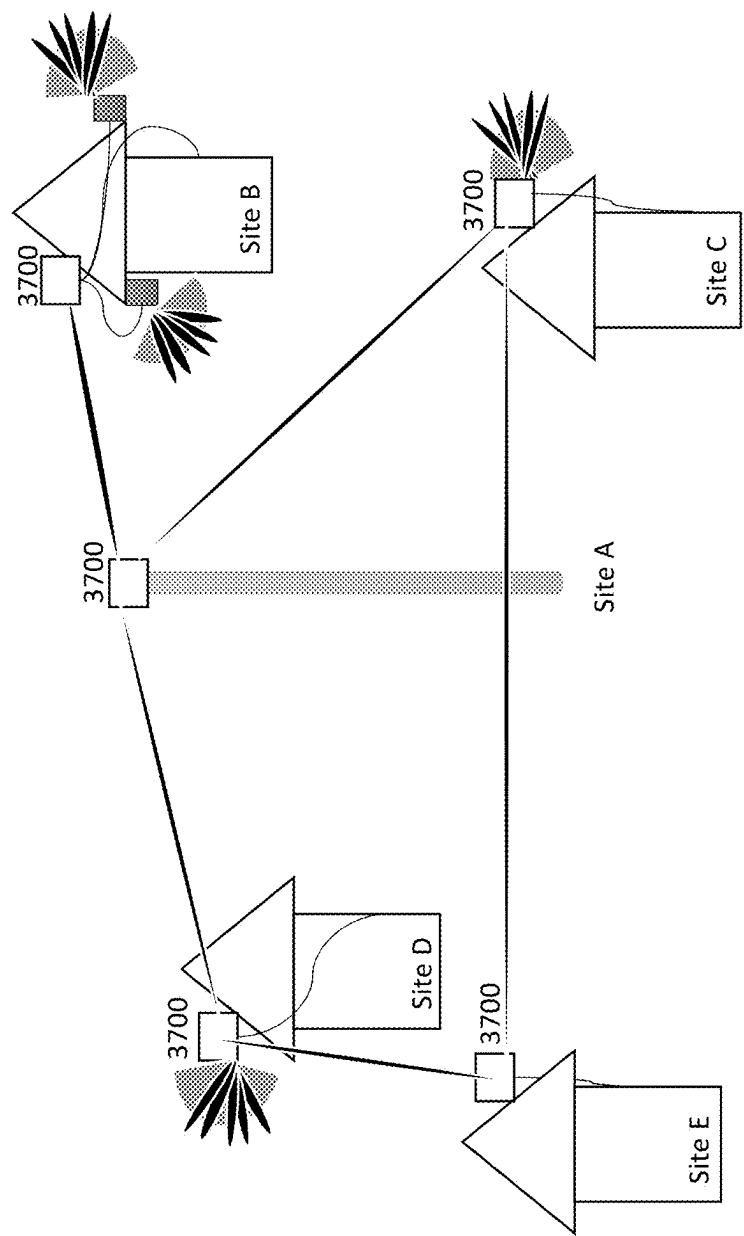


FIG. 24

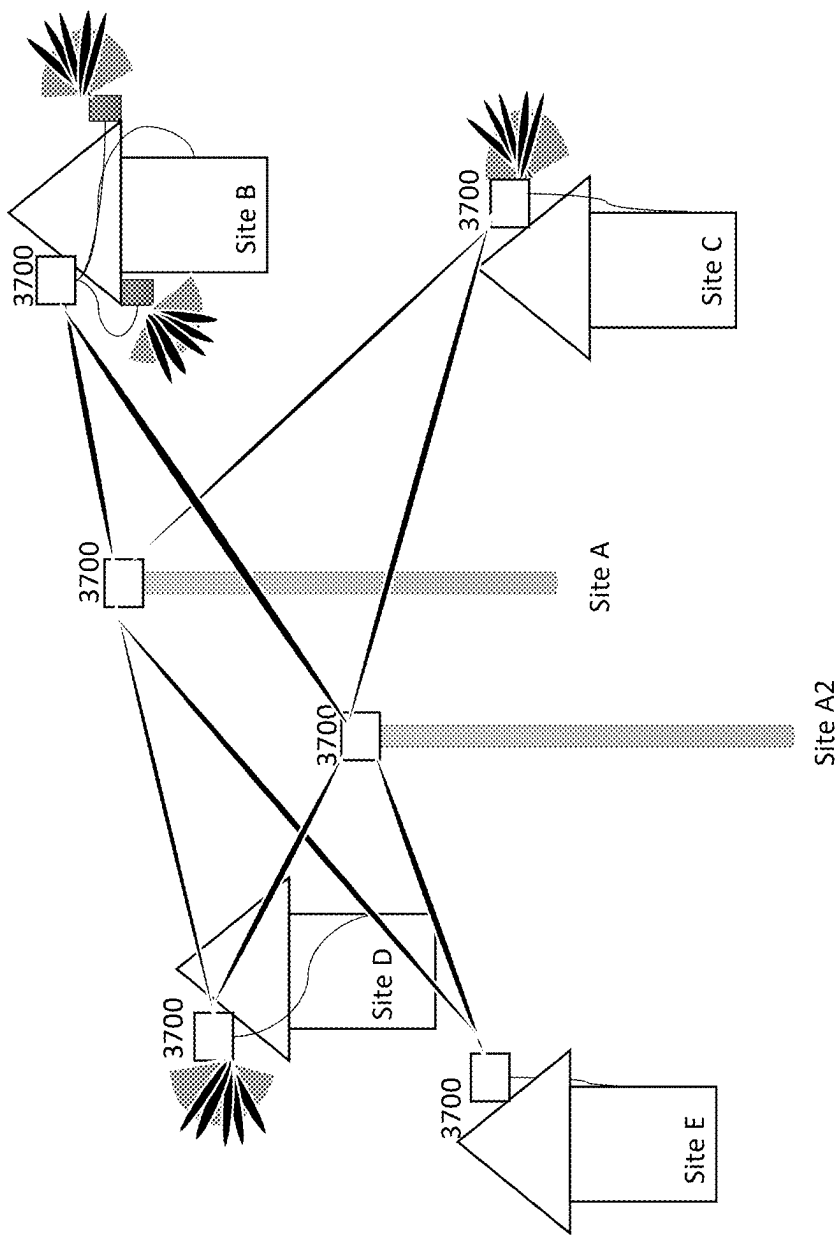


FIG. 25A

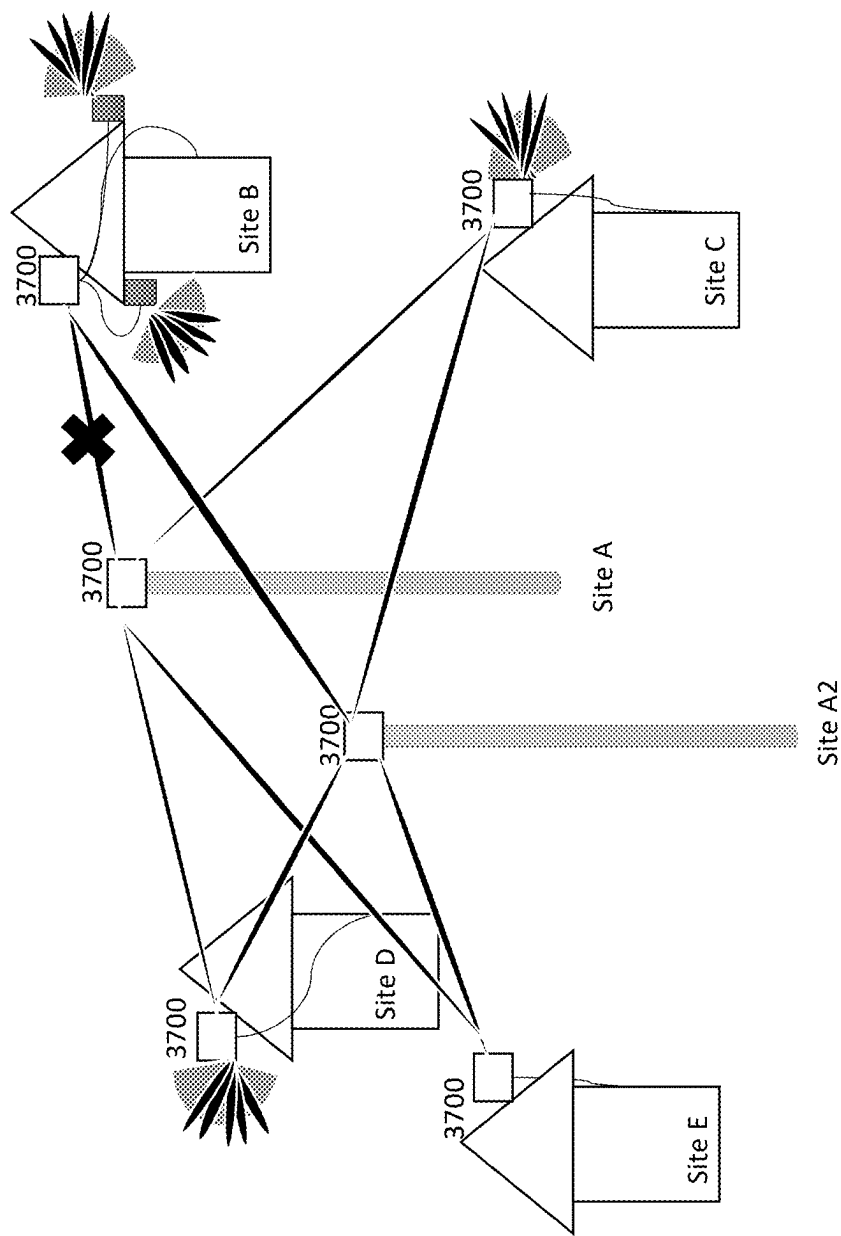


FIG. 25B

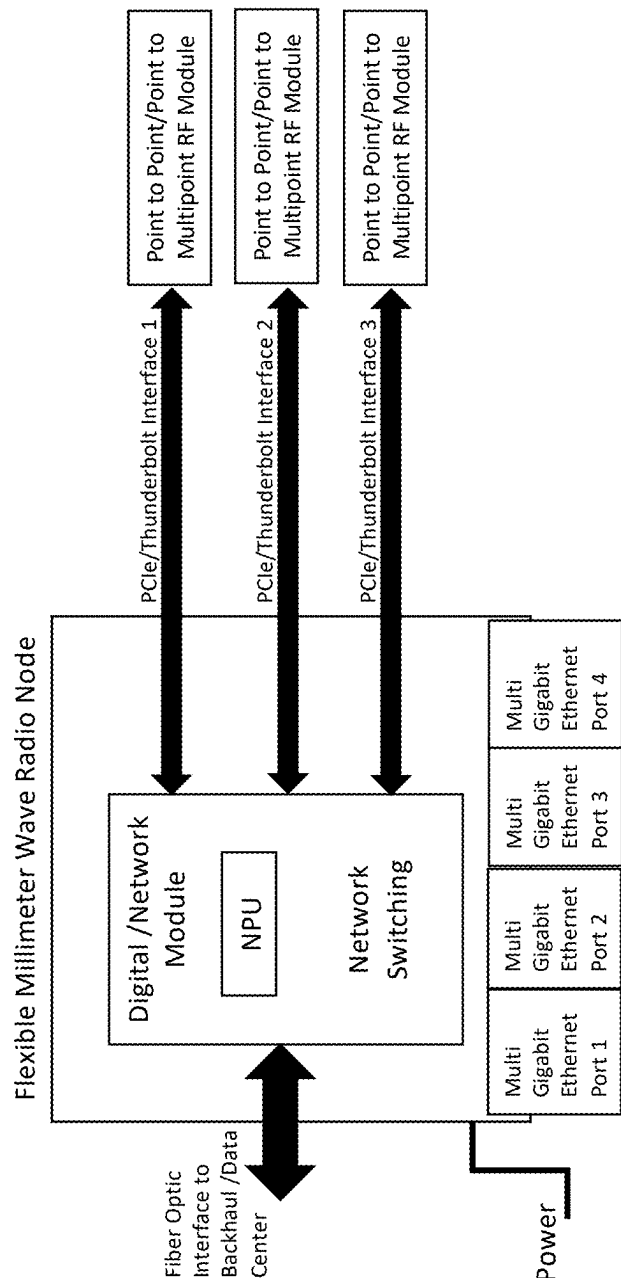


FIG. 26

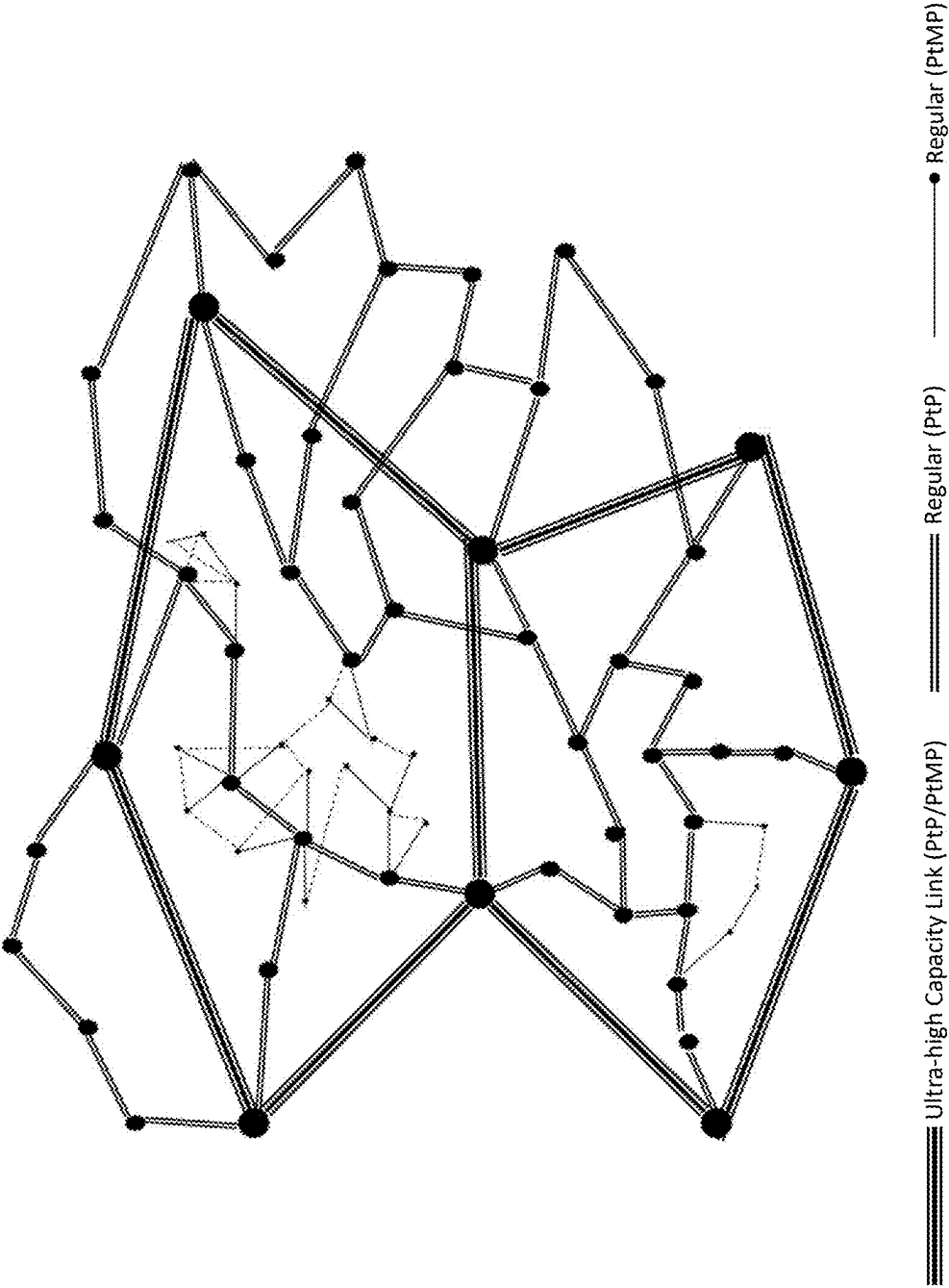


FIG. 27

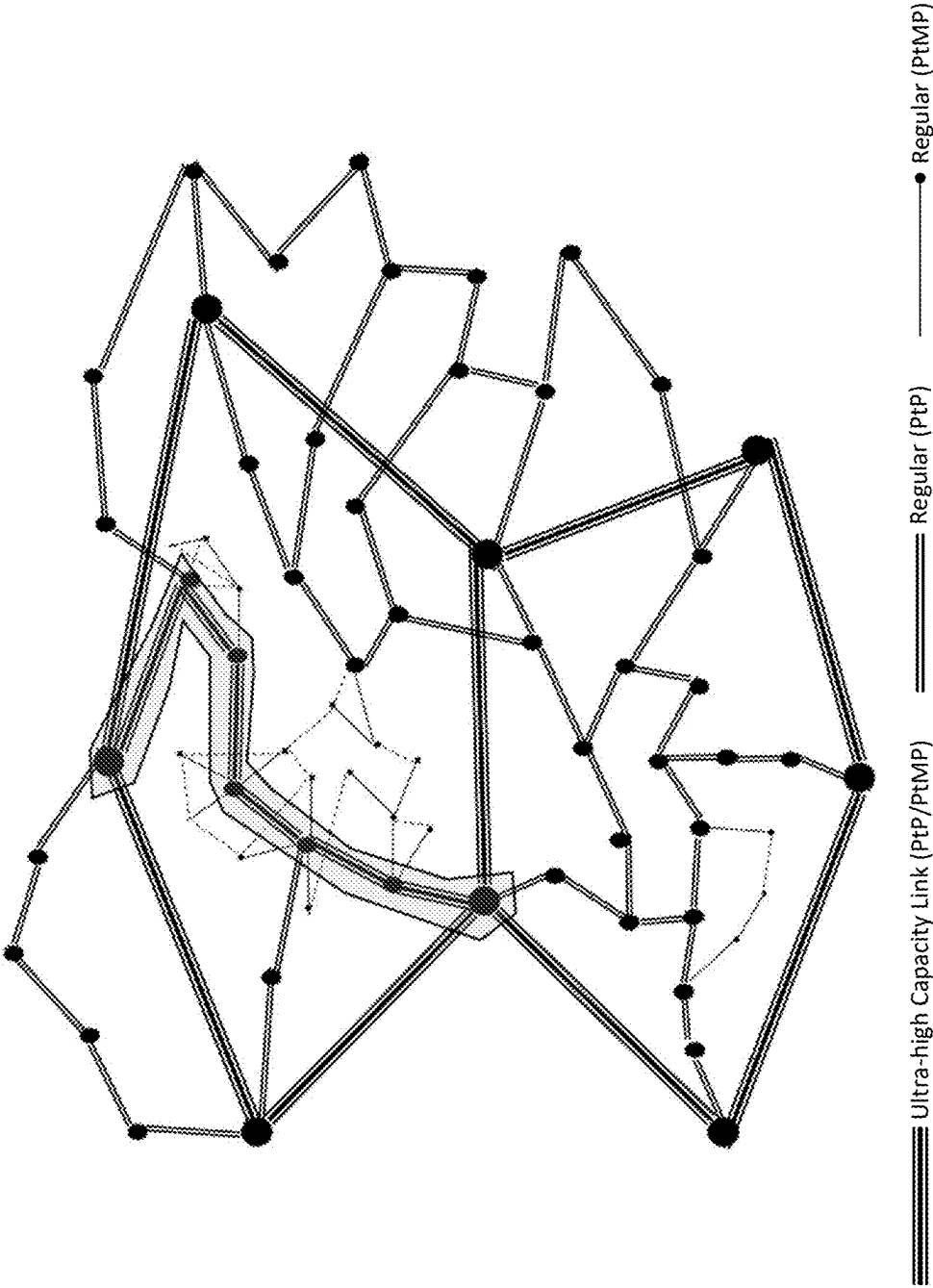


FIG. 28

SYSTEMS AND METHODS FOR IMPROVING WIRELESS MESH NETWORKS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of and claims priority to U.S. Non-Provisional patent application No. 18/479,145, filed Oct. 2, 2023, and entitled “Systems and Methods for Improving Wireless Mesh Networks,” which is a continuation of U.S. Non-Provisional patent application Ser. No. 17/665,790, issued as U.S. Pat. No. 11,778,678, filed Feb. 7, 2022, and entitled “Systems and Methods for Improving Wireless Mesh Networks,” which is a continuation of U.S. Non-Provisional patent application Ser. No. 17/345,914, issued as U.S. Pat. No. 11,246,180, filed on Jun. 11, 2021 and entitled “Systems And Methods For Improving Wireless Mesh Networks,” which is a continuation of U.S. Non-Provisional patent application Ser. No. 16/680,457, issued as U.S. Pat. No. 11,044,617, filed on Nov. 11, 2019, and entitled “Systems and Methods for Improving Wireless Mesh Networks,” which is a continuation-in-part of U.S. Non-Provisional patent application Ser. No. 16/590,217, issued as U.S. Pat. No. 11,102,834, filed on Oct. 1, 2019 and entitled “Systems and Methods for Improving Wireless Mesh Networks,” which is a continuation-in-part of U.S. Non-Provisional patent application Ser. No. 16/508,289, issued as U.S. Pat. No. 10,966,266, filed Jul. 10, 2019 and entitled “Systems and Methods for Improving Wireless Mesh Networks,” which claims priority to the following U.S. provisional applications: (i) U.S. Provisional App. No. 62/856,697, filed Jun. 3, 2019 and entitled “A METHOD FOR BUILDING WIRELESS MESH NETWORK NODES,” (ii) U.S. Provisional App. No. 62/833,485, filed Apr. 12, 2019 and entitled “A METHOD FOR BUILDING WIRELESS MESH NETWORK NODES,” (iii) U.S. Provisional App. No. 62/771,508, filed Nov. 26, 2018 and entitled “A METHOD FOR IMPROVING WIRELESS MESH NETWORK USING DIRECT OPTICAL-TO-RF AND DIRECT-RF-TO-OPTICAL CONVERSION MODULE,” (iv) U.S. Provisional App. No. 62/753,885, filed Oct. 31, 2018 and entitled “METHODS OF BUILDING 60 GHZ MESH BASED NETWORK INFRASTRUCTURE FOR BLOCKCHAIN TECHNOLOGY BASED PLATFORMS,” and (v) U.S. Provisional App. No. 62/696,688, filed Jul. 11, 2018 and entitled “METHODS FOR BUILDING WIRELESS MESH NETWORK.” The contents of each of the foregoing patent applications are incorporated herein by reference in their entireties for all purposes.

BACKGROUND

[0002] Wired and wireless networking and communications systems are widely deployed to provide various types of communication and functional features, including but not limited to those for high speed home internet, security and automation, and/or others. These systems may be capable of supporting communication with a user through a communication connection or a system management action.

[0003] Current wireless mesh networking systems exhibit many shortcomings, including failing to account for extra protection for point-to-point narrow beam wireless paths. Such paths are highly directional and work only under perfect line-of-sight or near line-of-sight conditions. Once the wireless mesh network is built, certain events such as

vegetation growth or loss of an intermediary node can impact the line-of-sight paths between the links. This can result in single or multiple link failures in the network.

[0004] Thus, there exists multiple needs in the art for improved systems and methods relating to wireless communication mesh network design and operation.

OVERVIEW

[0005] The present disclosure, for example, relates to wireless networks and communications including, but not limited to, broadband internet services to end user, security and/or automation systems, and more particularly to narrow beam mesh networking and related operations and techniques.

[0006] In accordance with the present disclosure, disclosed herein are systems and methods that relate to narrow beam mesh networks, associated systems, and operations relating to network communication, including, in some embodiments, adjustments and modifications. The present systems and methods may facilitate designing, operating and/or adjusting/modifying wireless networking communications. In some embodiments, the present systems and methods relate to and account for wireless communication nodes with capability of establishing point-to-point or point-to-multipoint narrow beam communication link, point-to-point/point-to-multipoint steerable narrow beam communication link, combination of point-to-point and point-to-multipoint communication links, among other things.

[0007] In some instances, one or multiple links in a wireless communication network can fail due to certain events, including but not limited to growth in vegetation, loss of a node due to various reasons that can change the line-of-sight (LOS) conditions required for communication between two points in the wireless mesh network. In the present disclosure, some or all the nodes in the mesh network can host a 4G (LTE, LTE Advanced, LTE Pro, WiMAX, WiFi, etc.) technology-based small cell (eNB) and an UE (user equipment)/CPE (customer premises equipment) in a single enclosure to provide a redundant communication path between two points in a mesh network in an event a direct or in-direct LOS path between two mesh nodes fail.

[0008] In another aspect of the present disclosure, a wireless mesh network may include ultra-high-capacity nodes that are capable of establishing ultra-high-capacity links (e.g., point-to-point or point-to-multipoint bi-directional communication links) using a millimeter wave spectrum, including but not limited to 28 Ghz, 39 Ghz, 37/42 Ghz, 60 Ghz (including V band), or E-band frequencies, as examples. The higher capacity and/or extended range of these ultra-high-capacity nodes/links may be achieved via various advanced signal processing techniques. Further, these ultra-high-capacity nodes/links may be used in conjunction with other types of point-to-point and/or point-to-multipoint links to build a multi-layer wireless mesh network.

[0009] The foregoing has outlined rather broadly the features and technical advantages of examples according to this disclosure so that the following detailed description may be better understood. Additional features and advantages will be described below. It should be understood that the specific examples disclosed herein may be readily utilized as a basis for modifying or designing other structures for carrying out the same operations disclosed herein. Characteristics of the

concepts disclosed herein including their organization and method of operation together with associated advantages will be better understood from the following description when considered in connection with the accompanying figures. It should be understood that the figures are provided for the purpose of illustration and description only.

[0010] One of ordinary skill in the art will appreciate these as well as numerous other aspects in reading the following disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] A further understanding of the nature and advantages the present disclosure may be realized by reference to the following drawings.

[0012] FIG. 1 depicts an example diagram relating to a wireless networking and communication system, in accordance with various aspects of this disclosure.

[0013] FIG. 2 depicts an example wireless communication node comprising a module, in accordance with various aspects of this disclosure.

[0014] FIG. 3 depicts an example antenna pattern of a module, in accordance with various aspects of this disclosure.

[0015] FIG. 4 depicts an example communication link between two wireless communication nodes, in accordance with various aspects of this disclosure.

[0016] FIG. 5 depicts example wireless communication nodes, in accordance with various aspects of this disclosure.

[0017] FIG. 6 depicts example wireless communication nodes that are communicatively coupled, in accordance with various aspects of this disclosure.

[0018] FIG. 7 depicts another example wireless communication node comprising a module, in accordance with various aspects of this disclosure.

[0019] FIG. 8 depicts yet another example wireless communication node comprising a module, in accordance with various aspects of this disclosure.

[0020] FIG. 9 depicts an example wireless communication node comprising two modules, in accordance with various aspects of this disclosure.

[0021] FIG. 10 depicts another example wireless communication node comprising two modules, in accordance with various aspects of this disclosure.

[0022] FIG. 11A depicts an example wireless communication node at a given time that can dynamically change type of links between wireless communication nodes, in accordance with various aspects of this disclosure.

[0023] FIG. 11B depicts an example wireless communication node at another given time that can dynamically change type of links between wireless communication nodes, in accordance with various aspects of this disclosure.

[0024] FIG. 12 depicts an example location of a communication network that can be a seed or an anchor home, in accordance with various aspects of this disclosure.

[0025] FIG. 13 depicts another example location of a communication network that can be a seed or an anchor home, in accordance with various aspects of this disclosure.

[0026] FIG. 14 depicts another example wireless communication node comprising a module, in accordance with various aspects of this disclosure.

[0027] FIG. 15 depicts an example of multiple modules connected to a tower, in accordance with various aspects of this disclosure.

[0028] FIG. 16 depicts another example diagram relating to a wireless networking and communication system, in accordance with various aspects of this disclosure.

[0029] FIG. 17 depicts yet another example diagram relating to a wireless networking and communication system, in accordance with various aspects of this disclosure.

[0030] FIG. 18 depicts still another example diagram relating to a wireless networking and communication system, in accordance with various aspects of this disclosure.

[0031] FIG. 19 depicts an example communication module based on direct RF-to-Optical and direct Optical-to-RF conversion, in accordance with various aspects of this disclosure.

[0032] FIG. 20 depicts an example router/switch, in accordance with various aspects of this disclosure.

[0033] FIG. 21 depicts an example block diagram of a communication module, in accordance with various aspects of this disclosure.

[0034] FIG. 22 depicts an example block diagram of a ptmp radio module of a communication node, in accordance with various aspects of this disclosure.

[0035] FIG. 23 depicts an example block diagram of a ptp radio module of a communication node, in accordance with various aspects of this disclosure.

[0036] FIG. 24 depicts an example mesh network comprising a plurality of communication nodes, in accordance with various aspects of this disclosure.

[0037] FIG. 25A depicts another example mesh network comprising a plurality of communication nodes, in accordance with various aspects of this disclosure.

[0038] FIG. 25B depicts yet another example mesh network comprising a plurality of communication nodes, in accordance with various aspects of this disclosure.

[0039] FIG. 26 depicts an example modified version of a flexible millimeter wave radio box, in accordance with various aspects of this disclosure.

[0040] FIG. 27 depicts an example of a multi-layer wireless mesh network, in accordance with various aspects of this disclosure.

[0041] FIG. 28 depicts another example of a multi-layer wireless mesh network, in accordance with various aspects of this disclosure.

DETAILED DESCRIPTION

[0042] Current wireless mesh networking systems exhibit many shortcomings, including failing to account for extra protection for a high reliability wireless path for carrying backhaul data that carries control signaling data along with user data for users in the network segment. Current wireless mesh networking systems use the same or similar beam transmission techniques for an access path that carries data for a single user and backhaul path that may affect network performance as backhaul paths tend to be more sensitive to interference and other signal inhibitors and can degrade the performance of entire network segment. Thus, there exists multiple needs in the art for improved systems and methods relating to wireless communication network design and operation.

[0043] Referring to FIG. 1, a data communication network capable of providing multigigabit internet speeds through wireless point-to-point and point-to-multipoint links is illustrated. Communication network 100 in FIG. 1 includes Tower/fiber access points 101 and 102. Tower/fiber access points 101 and 102 can be co-located or can be located at

different physical locations. Tower/fiber access points **101** and **102** have access to a high bandwidth dark fiber capable of providing up to several hundred gigabits/second of data throughput. Tower/fiber access points **101** and **102** provide backhaul connectivity between a core network/data center (not shown in the FIG. 1 for the sake of simplicity) and a seed home of the communication network described below. Tower/Fiber access points **101** and **102** also host wireless point-to-point communication nodes **121** and **124**.

[0044] As shown, wireless communication nodes **121** and **124** are capable of reception and transmission of high bandwidth (multiple gigahertz) signals operating at very high frequency (6 Ghz~100 Ghz such as 28 Ghz, V band, E band, etc.). Wireless communication nodes **121** and **124** each comprise a baseband/digital unit equipped with components including but not limited to a processor, memory, etc. Wireless communication nodes **121** and **124** also each comprise an RF unit and antenna unit. The antenna subsystem of wireless communication nodes **121** and **124** is capable of reception and transmission of directional signals where significant portion of the signal energy is concentrated within a few degrees around the antenna boresight (e.g., within a range of 0.5 degrees to 5 degrees), both in vertical and horizontal directions, in contrast to omni directional antennas where signal energy is close to evenly spread across 360° degrees.

[0045] As further shown in FIG. 1, Communication network **100** includes seed homes **111** and **115**. Examples of seed homes include detached single-family homes, multi-dwelling units (MDUs), small/medium business (SMB), etc., where communication equipment nodes can be deployed on rooftops. Seed homes **111** and **115** host wireless point-to-point communication nodes **122** and **123**. Wireless communication nodes **122** and **123** are capable of reception and transmission of high bandwidth (multiple gigahertz) signals operating at very high frequency (6 Ghz~100 Ghz such as 28 Ghz, V band, E band, etc.).

[0046] Wireless communication nodes **122** and **123** comprise a baseband/digital unit equipped with components including but not limited to a processor, memory, etc. Wireless communication nodes **122** and **123** also comprise an RF unit and antenna unit. The antenna subsystem of wireless communication nodes **122** and **123** is capable of reception and transmission of directional signals where a significant portion of the signal energy is concentrated within few degrees around the antenna boresight (e.g., within a range of 0.5 degrees to 5 degrees), both in vertical and horizontal directions, in contrast to omni directional antennas where signal energy is close to evenly spread across 360° degrees.

[0047] Wireless communication node **121** residing in Tower/fiber access point location **101** and wireless communication node **122** residing in seed home **111** work together to form a bi-directional high bandwidth communication point-to-point data link **141** that provides connectivity between Tower/fiber access point **101** and seed home **111** segment of communication network **100**. Similarly, wireless communication node **124** residing in Tower/fiber access point location **102** and wireless communication node **123** residing in seed home **115** work together to form a bi-directional high bandwidth communication point-to-point data link **142** that provides connectivity between Tower/fiber access point **102** and seed home **115** segment of the communication network **100**.

[0048] Seed home **111**, in addition to wireless communication node **122**, hosts a second wireless communication node **131**. Second wireless communication node **131** comprises multiple independent transmission/reception modules. Each module of the wireless communication node **131** is capable of reception and transmission of high bandwidth (multiple gigahertz) signals operating at very high frequency (6 Ghz~100 Ghz such as 28 Ghz, V band, E band, etc.). Each module of wireless communication node **131** comprises an independent baseband/digital unit equipped with components including but not limited to a processor, memory, etc. Each module in wireless communication node **131** also comprises an independent RF unit and independent antenna unit. The antenna subsystem of each wireless communication node **131**'s module is capable of reception and transmission of directional signals where significant portion of the signal energy is concentrated within few degrees around the antenna boresight (e.g., within a range of 0.5 degrees to 5 degrees), both in vertical and horizontal directions, in contrast to omni directional antennas where signal energy is close to evenly spread across 360° degrees.

[0049] Communication network **100** also includes multiple anchor homes **112**, **113** and **114**. Each of these anchor homes host a wireless communication node similar to wireless communication node **131** described above. However, unlike seed homes, a wireless communication node on an anchor home only provides wireless connectivity to one or more anchor homes and/or seed homes but does not provide connectivity to the Tower/Fiber access point. For example, anchor home **112** hosts wireless communication node **132**. A first module of wireless communication node **132** residing in anchor home **112** and another module of wireless communication node **131** residing in seed home **111** work together to form a bi-directional high bandwidth communication point-to-point data link **151** that provides connectivity between seed home **111** and anchor home **112** segment of the communication network **100**. Similarly, as another example, a second module of wireless communication node **132** residing in anchor home **112** and a module of wireless communication node **133** residing in anchor home **113** work together to form a bi-directional high bandwidth communication point-to-point data link **153** that provides connectivity between anchor home **112** and anchor home **113**. As yet another example, a third module of wireless communication node **132** residing in anchor home **112** and a module of wireless communication node **135** residing in seed home **115** work together to form a bi-directional high bandwidth communication point-to-point data link **154** that provides connectivity between anchor home **112** and seed home **115**. As a further example, another module of wireless communication node **131** residing in seed home **111** and a module of wireless communication node **134** residing in anchor home **114** work together to form a bi-directional high bandwidth communication point-to-point data link **152** that provides connectivity between anchor home **114** and seed home **111**. As another example, another module of wireless communication node **134** residing in anchor home **114** and a module of wireless communication node **135** residing in seed home **115** work together to form a bi-directional high bandwidth communication point-to-point data link **156** that provides connectivity between anchor home **114** and seed home **115**. Other examples are possible as well.

[0050] Bi-directional communication links **141**, **142**, **151**, **152**, **153**, **154** & **155** shown in FIG. 1 can use various

different multiple access schemes for transmission and reception including but not limited to frequency division multiple access (FDMA), time division multiple access (TDMA), single carrier FDMA (SC-FDMA), single carrier TDMA (SC-TDMA), code division multiple access (CDMA), orthogonal frequency division multiple access (OFDMA), and/or non-orthogonal multiple access (NOMA) as described in various generations of communication technologies including 1G, 2G, 3G, 4G, 5G and 6G, etc. Bi-directional communication links **141**, **142**, **151**, **152**, **153**, **154** & **155** formed by a set of communication nodes comprising two or more of **121**, **122**, **123**, **124**, **131**, **132**, **133**, **134**, and/or **135** are capable of data information transfer via a variety of digital transmission schemes, including but not limited to amplitude modulation (AM), phase modulation (PM), pulse amplitude modulation/quadrature amplitude modulation (PAM/QAM), and/or ultra-wide band (UWB) pulse modulation (pico-second pulses), etc.

[0051] In FIG. 1, two Tower/fiber access points **101** & **102**, two seed homes **111** & **115** and three anchor homes **112**, **113** & **114** and seven bi-directional point to point data links **141**, **142**, **151**, **152**, **153**, **154** & **155** are shown to illustrate an example of a communication network. However, in general, it should be understood that communication network **100** can include a different number of Tower/fiber nodes, seed homes, anchor homes and/or communication links, which may depend on the specific layout of a particular instantiation of the communication network deployed in the field. Similarly, although, FIG. 1 shows four communication nodes **121**, **122**, **123** & **124** that provide connectivity between a Tower/fiber access point (e.g., Tower/fiber access points **101**, **102**) and a seed home, five communication nodes **131**, **132**, **133**, **134** & **135** that provide connectivity between two anchor homes or between an anchor and a seed home, the number of these communication nodes can vary from one communication network to another communication network, which may depend on the specific size and layout of a particular instantiation of the communication network. It should also be understood that communication network **100** may also contain other nodes (e.g., network switches/routers, etc.) that are omitted here for the sake of simplicity.

[0052] Referring to FIG. 2, one possible embodiment of wireless communication node **131** of FIG. 1 is shown as wireless communication node **200**. Wireless communication node **200** in FIG. 2 comprises a module labelled as “Module A.” As shown, Module A comprises a base band unit or digital unit **201** which runs the physical layer level protocol including digital modulation/demodulation (modem) and other higher layer protocols such as MAC layer etc. Base band unit **201** interacts with other nodes of communication network that are external to the wireless communication node **200** via a wired medium.

[0053] Module A also includes RF unit **202** which, among other things, performs processing of intermediate frequency (IF) signals and defines the frequency range of the radio signals that can be transmitted or received via Module A. RF unit **202** is capable of operating over a wide range of frequencies (e.g., V band frequencies ranging from 57 Ghz to 71 Ghz).

[0054] Further, as shown, Module A also comprises antenna unit **203** which performs the transmission and reception of over the air radio signals. Antenna unit **203** is capable of transmitting and receiving extremely narrow

beam of signals. Antenna unit **203** may be constructed with metamaterials that have excellent properties of controlling the directionality of radio signals that cannot be exhibited by ordinary antennas. Module A with the help of antenna unit **203** is capable of establishing point-to-point links with a different module residing in a different wireless communication node.

[0055] Referring to FIG. 3, an example of an antenna pattern of Module A created by antenna unit **203** is shown. It can be seen from the antenna pattern in FIG. 3 that the beam width of antenna unit **203** of Module A is extremely narrow (less than a degree) and the side lobe power levels start to drop at a rapid rate. For instance, as shown, approximately 5-6 degrees from the main lobe, power levels may drop by more than 30 dB.

[0056] It should be understood that the antenna pattern of antenna unit **203** shown in FIG. 3 is just one example showing the extremely narrow beam antenna pattern generation capability of Module A. In other instances, due to change in antenna elements, size, frequency, etc., different patterns may be generated. Further, while Module A can be constructed using metamaterials described above, it should be understood that Module A can be constructed using a parabolic antenna or other types of antennas. However, it should be understood that the main characteristic of generation of extremely narrow antenna beam pattern is common to all the instances of Module A.

[0057] Referring to FIG. 4, a point-to-point wireless communication link **400** established between two wireless communication nodes **401** and **402** is shown. Wireless communication nodes **401** and **402** each host a single communication module (i.e., “Module A”) that may take the form similar to Module A depicted in FIG. 2 and described above. As shown in FIG. 4, due to the antenna unit characteristics of Module A in wireless communication node **401** and **402**, the bi-directional point-to-point link **400** may have an extremely narrow beam. This transmission and reception capability of radio signals over an extremely narrow beam via point-to-point link **400** provides interference immunity in scenarios where there are a large number of wireless communication links established by multiple wireless communication nodes concentrated in a small area and operating in the same frequency.

[0058] In some implementations, Module A can additionally provide beam steerability characteristics in addition to the capability of transmitting and receiving data over extremely narrow beams as explained above and illustrated in the context of FIGS. 2-4.

[0059] For example, referring to FIG. 5, a wireless communication node **501** comprising Module A, a second wireless communication node **502** comprising Module A and a third wireless communication node **503** comprising Module A is shown. During time T1, Module A of wireless communication node **501** and Module A of wireless communication node **502** work together to establish an extremely narrow beam based bi-directional link **500** for the exchange of information data between wireless communication nodes **501** and **502**. Due to some trigger, Module A of wireless communication node **501** may invoke the beam steering capability of the module and change the direction of the antenna transmission and reception beam towards wireless communication node **503** and work together with Module A of wireless communication node **503** to dynamically establish a bi-directional extremely narrow beam-based link **500**

between wireless communication node **501** and wireless communication node **503** during time T2. The trigger for this beam steering can be due to changes in the link condition between wireless communication node **501** and wireless communication node **502** which may involve various factors, including but not limited to a change from a LOS path to a non-LOS path due to a change in environment, increased interference, a change in position of wireless communication node **502** with respect to wireless communication node **501**, and/or instructions from higher layers, etc.

[0060] In one embodiment, wireless communication node **503** can be different than wireless communication node **502**. In another embodiment, wireless communication node **503** can be the same as wireless communication node **502** but in a different physical location.

[0061] In some embodiments, wireless communication nodes defined above and discussed in the context of FIGS. 2-5 can host more than one module. This allows a wireless communication node to communicate simultaneously with multiple wireless communication nodes by establishing multiple extremely narrow beam bi-directional links with the help of multiple modules (e.g., multiple Module As) belonging to different wireless communication nodes working together.

[0062] As one example to illustrate, referring to FIG. 6, wireless communication nodes **601** and **602** each host two Module As labeled “1” and “2,” while wireless communication nodes **603** and **604** each host a single Module A. As shown, a 1st Module A of wireless communication node **601** and a 1st Module A of wireless communication node **602** work together to establish extremely narrow bi-directional beam-based link **600** to provide wireless connection between wireless communication node **601** and **602**. Similarly, a 2nd Module A of wireless communication node **601** and **602** and a 1st (and only) Module A of wireless communication node **603** and **604** respectively work together to establish extremely narrow bi-directional beam-based links **610** and **620** to provide wireless connection between wireless communication nodes **601-603** and **602-604** respectively.

[0063] In one embodiment, the 1st and 2nd Module A of wireless communication nodes **601** and **602** can be inside the same physical enclosure and in other embodiments, the 1st Module A of wireless communication nodes **601** and **603** can be inside one physical enclosure and the 2nd Module A of wireless communication nodes **601** and **603** can be inside a different physical enclosure. In embodiments where different Module As belonging to the same wireless communication node are contained in separate physical enclosures, these Module As can be connected via a wired link as they are co-located in the same seed home or anchor home.

[0064] In FIG. 6, a maximum of two Module As are shown to be contained in a wireless communication node that enables the wireless communication node to establish two independent bi-directional links with different wireless communication nodes simultaneously. However, it should be understood that a wireless communication node can host more than two Module As and the maximum number of Module As that a wireless communication node can host may depend on the maximum total power available to the wireless communication node.

[0065] Further, it should be understood that in one embodiment, all Module As belonging to the same wireless

communication node may operate on the same carrier frequencies of a frequency band, and in other embodiments, different Module As belonging to same wireless communication node may operate on different carrier frequencies of a frequency band.

[0066] Referring to FIG. 7, another embodiment of wireless communication node **131** of FIG. 1 is shown as wireless communication node **700**. Wireless communication node **700** in FIG. 7 comprises a single module labeled as “Module B.” Module B comprises base band unit or digital unit **701** which runs the physical layer level protocol including digital modulation/demodulation (modem) and other higher layer protocols such as MAC layer, etc. Base band unit **701** interacts with other nodes of the communication network that are external to the wireless communication node **700** via wired medium.

[0067] Module B also includes RF unit **702**, which among other things processes intermediate frequency (IF) signals and defines the frequency range of the radio signals that can be transmitted or received with Module B. RF unit **702** is capable of operating over a wide range of frequencies (e.g., V band frequencies ranging from 57 GHz to 71 GHz).

[0068] Further, Module B comprises antenna unit **703**, which performs the transmission and reception of over the air radio signals. Antenna unit **703** may be an active antenna system (AAS) that comprises a phased array of transmitters and receivers that are capable of beamforming and creating multiple beams simultaneously in different directions. By virtue of the simultaneous creation of multiple beams in different directions, AAS of antenna unit **703** enables the wireless communication node **700** to establish point-to-multipoint wireless communication links with multiple wireless communication nodes. Hence Module B with the help of antenna unit **703** is capable of establishing point-to-multipoint links with a different module residing in a different wireless communication node.

[0069] As further shown in FIG. 7, Module B residing in wireless communication node **700** is shown to create 1 to N multiple beams with the help of AAS of antenna unit **703**. Value N depends on the number of transmit and receive antennas in AAS of antenna unit **703**. Specifically, it can be seen that wireless communication unit **700** is connected to wireless communication unit **710**, wireless communication unit **720**, and wireless communication unit **7N0** via bi-directional beam **1**, beam **2** and beam **N** respectively. It can also be seen from the antenna pattern in FIG. 7 that the beam width of point-to-multipoint beams of antenna unit **703** of Module B are not extremely narrow (e.g. 3 dB beam width of 7~10 degree) and side lobes power levels do not start to drop at a rapid rate which is in contrast to the antenna pattern of the antenna unit belonging to Module A described above and discussed in the context of FIGS. 2-6.

[0070] Further, Module B of wireless communication node **700** also differs from Module A (discussed above in the context of FIGS. 2-6) with respect to the limitation that the multiple bi-directional links operate in a single frequency range at a given time. For example, signal beams **1** to **N** that connect wireless communication node **700** to wireless communication nodes **710** to **7N0** respectively may only operate within the same frequency range at a given instant of time. It is to be noted that at a different instant, all beams **1** to **N** can switch to operate at a frequency range different from the frequency range used in the previous time instant, however, frequency range of an individual beam remains the same as

the frequency range of all the other N-1 beams at a given instant of time. Hence, with respect to Module B, although due to phased antenna arrays can create multiple beams to create point-to-multi point links to connect one wireless communication node with multiple wireless communication nodes as shown in FIG. 7, an interference profile at the receiver side with such point-to-multipoint network is inferior to an interference profile of point-to-multipoint network where a wireless communication node hosts multiple Module As and creates multiple point-to-point links as shown in FIG. 6, where wireless communication node 601 uses two Module As to connect to wireless communication node 602 and wireless communication node 603 simultaneously. The main reasons of high interference with Module B may be due to (1) individual phased antenna array-based beams that are not as narrow as extremely narrow beams generated by metamaterial-based antenna of Module A and/or (2) all beams of Module B belonging to one wireless communication unit that cannot operate at different frequency ranges unlike multiple point-to-point narrow beams of wireless communication node that host multiple Module As.

[0071] Referring to FIG. 8, another possible embodiment of wireless communication node 131 of FIG. 1 is shown as wireless communication node 800 and wireless communication node 810. Wireless communication node 800 in FIG. 8 comprises a module labeled as "Module C." Module C comprises a base band unit or digital unit which runs the physical layer level protocol including digital modulation/demodulation (modem) and other higher layer protocols such as MAC layer etc. Module C's baseband unit interacts with other nodes of a communication network that are external to the wireless communication node 800 via wired medium.

[0072] Module C also includes an ultra-wide band antenna embedded with the baseband unit. Module C is capable of generation, transmission, and reception of extremely short duration pulses (few pico seconds long) and uses pulse modulation (and its variations such as pulse amplitude modulation, etc.) to transmit data at extremely high rates (e.g., greater than 100 Gbps) by transmitting signals over a very wide range of frequencies. In one embodiment, pulses used for communication by Module C can use frequencies ranging from few hundred megahertz to few hundred gigahertz. One of ordinary skill in the art will appreciate that the range of frequencies used by pulses generated by Module C of wireless communication unit 800 can take a different range as well. Moreover, multiple module Cs can be placed together to create a 1, 2, or 3 dimensional array. Elements of this array (e.g., module C) are capable of performing a time synchronized transmission for beam forming. This allows the RF signal energy of the Pico second/UWB pulses to focus in a desired (receiver) direction and can also enable the creation of null or low RF signal energy of the Pico second/UWB pulse in other directions to avoid interference.

[0073] One fundamental difference between the characteristic of signals generated by Module C and signals generated by Module A and/or Module B is that these signals generated by Module C are ultra wide band (UWB) signals and their power spectral density over the entire range of frequencies is very low. In this respect, these UWB signals do not create interference with other signals operating on a narrow band of frequencies as these UWB signals are treated as noise by receivers of normal wireless communication nodes.

[0074] As further shown in FIG. 8, Module C of wireless communication node 800 and Module C of wireless communication unit 810 establish a link 801 by working together. As explained above, such a communication link 801 operates over an ultra-wide range of frequencies. However, even in the presence of other wireless communication nodes (not shown in FIG. 8) such as wireless communication nodes with Module A or Module B that operate on a narrow band of frequencies compared to Module C of wireless communication node 800, performance of network is not impacted as power spectral density over the frequency range of communication link 801 that overlaps with frequency ranges on which a nearby wireless communication node using narrow band signals using for example Module A and/or Module B operates is very low and is treated as noise by the receivers of Module A and/or Module B.

[0075] In another preferred embodiment, in line with the discussion above, wireless communication node 131 in FIG. 1 can host two types of modules. This allows a wireless communication node to communicate simultaneously with multiple wireless communication nodes and with two different interference profiles.

[0076] As one example to illustrate, referring to FIG. 9, wireless communication node 910 hosts one Module A and one Module B. Module A of wireless communication node 910 and a communication module of wireless communication node 920 work together to establish extremely narrow bi-directional beam-based link 901 to provide wireless connection between wireless communication nodes 910 and 920. Simultaneously, Module B of wireless communication node 910 which is based on AAS and generates multiple beams simultaneously creates a point-to-multipoint link that connects wireless communication node 910 with wireless communication nodes 930, 940, 950 and 960. Specifically, Module B of wireless communication node 910 coordinates with (1) a module of wireless communication node 930 to establish bi-directional beam 902, (2) a module of wireless communication node 940 to establish bi-directional beam 903, (3) a module of wireless communication node 950 to establish bi-directional beam 904, and (4) a module of wireless communication node 960 to establish bi-directional beam 905. In one embodiment, extremely narrow beam 901 and group of beams including 902, 903, 904 and 905 may all operate within the same range of carrier frequencies at a given time. In another embodiment, extremely narrow beam 901 may operate within a different range of frequencies compared to the range of frequencies used by the group of beams including 902, 903, 904 and 905 at a given time.

[0077] In one embodiment, Module A and Module B of wireless communication node 910 can be inside the same physical enclosure. In other embodiments, Module A and Module B of wireless communication node 910 can be inside two separate physical enclosures. In such embodiments where Module A and Module B belong to the same wireless communication node contained in separate physical enclosures, Module A and Module B can be connected via a wired link as they are co-located in the same seed home or anchor home.

[0078] In FIG. 9, a maximum of two modules (i.e., a single Module A and a single Module B) are shown to be contained in a wireless communication node 910 that enables the wireless communication node to establish two independent and different types of bi-directional links with different wireless communication nodes simultaneously. However, it

should be understood that wireless communication node **910** can host more than two modules (e.g., a combination of one or more Module As and one or more Module Bs) and the maximum number of total modules that a wireless communication node can host may depend on the maximum total power available to the wireless communication node. Further, it should be understood that in one embodiment, all modules belonging to same wireless communication node may operate on the same carrier frequencies of a frequency band but in other embodiments, different modules belonging to the same wireless communication node may operate on different carrier frequencies of a frequency band.

[0079] As noted above, wireless communication nodes **131** in FIG. 1 can host more than one type of module. This allows a wireless communication node to communicate simultaneously with multiple wireless communication nodes and with different interference profiles.

[0080] As another example to illustrate, referring to FIG. 10, wireless communication node **1010** hosts one Module C and one Module B. Module C of wireless communication node **1010** and Module C of wireless communication node **1020** work together to establish extremely high data rate ultra-wide frequency and low power spectral density beam-based link **1001** to provide wireless connection between wireless communication nodes **1010** and **1020**. Module B of wireless communication node **1010**, which is based on AAS and generates multiple beams simultaneously, creates a point-to-multipoint link that connects wireless communication node **1010** with wireless communication nodes **1030**, **1040**, **1050** and **1060**. Specifically, Module B of wireless communication node **1010** coordinates with (1) a module of wireless communication node **1030** to establish bi-directional beam **1002**, (2) a module of wireless communication node **1040** to establish bi-directional beam **1003**, (3) a module of wireless communication node **1050** to establish bi-directional beam **1004**, and (4) a module of wireless communication node **1060** to establish bi-directional beam **1005**.

[0081] In one embodiment, Module C and Module B of wireless communication node **1010** can be inside same physical enclosure. In other embodiments, Module C and Module B of wireless communication node **1010** can be inside two separate physical enclosures. In such an embodiment where Module C and Module B belong to the same wireless communication node contained in separate physical enclosures, Module C and Module B can be connected via a wired link as they are co-located in same seed home or anchor home.

[0082] In FIG. 10, a maximum of two modules (i.e., a single Module C and a single Module B) are shown to be contained in a wireless communication node **1010** that enables the wireless communication node to establish two independent and different types of bi-directional links with different wireless communication nodes simultaneously. However, it should be understood that wireless communication node **1010** can host more than two types of module (e.g., a combination of Module A, Module B and/or Module C) and the maximum number of total modules that a wireless communication node can host may depend on the maximum total power available to the wireless communication node. It should be also understood that in one embodiment, all modules belonging to same wireless communication node may operate on same carrier frequencies of a frequency band, while in other embodiments, different modules

belonging to same wireless communication node may operate on different carrier frequencies of a frequency band.

[0083] In another preferred embodiment, wireless communication nodes **131** in FIG. 1 can host more than one type of module and dynamically change the type of link between wireless communication nodes. This allows a wireless communication node to communicate simultaneously with multiple wireless communication nodes and with different interference profiles and to adapt with changes in network environment.

[0084] As one example to illustrate, referring to FIG. 11A, wireless communication node **1110** hosts a Module C or Module A along with a Module B. During time T1, Module A/Module C of wireless communication node **1110** and communication module of wireless communication node **1010** work together to establish either an extremely high data rate ultra-wide frequency low power spectral density beam or extremely narrow beam-based link **1101** to provide wireless connection between wireless communication nodes **1110** and **1120**. At substantially the same time duration T1, Module B of wireless communication node **1110** which is based on AAS and generates multiple beams simultaneously creates a point-to-multipoint link that connects wireless communication node **1110** with wireless communication nodes **1130**, **1140**, **1150** and **1160**. Specifically, Module B of wireless communication node **1110** coordinates with (1) a module of wireless communication node **1130** to establish bi-directional beam **1102**, (2) a module of wireless communication node **1140** to establish bi-directional beam **1103**, (3) a module of wireless communication node **1150** to establish bi-directional beam **1104**, and (4) a module of wireless communication node **1160** to establish bi-directional beam **1105**.

[0085] Referring to FIG. 11B, at a different time T2, due to some trigger, Module A/Module C of wireless communication node **1110** may dynamically switch its wireless link from wireless communication node **1120** to wireless communication node **1140** by steering the beam towards wireless communication node **1140**. At the same time or after receiving instructions from a higher layer, Module B of wireless communication node **1110** disconnects its link with wireless communication node **1140** via beam **1103** and generates a new beam **1113** in the direction of wireless communication node **1120** and establishes connection with wireless communication node **1120**. Trigger for this beam steering can be due to changes in the link condition between wireless communication node **1110** and wireless communication node **1120** or **1140**, which may involve various factors, including but not limited to change from a LOS path to a non-LOS path due to a change in environment, increased interference, a change in position of wireless communication node **1120** or **1140** with respect to wireless communication node **1110**, instructions from higher layers, etc.

[0086] As shown in FIGS. 11A-B, dynamic link switching may occur between wireless communication nodes **1110**, **1120** and **1140**. However, it should be understood that dynamic switching can also occur between different communication nodes.

[0087] In some instances, one or more wireless communication nodes may leave the communication network. In such case, links between nodes may be dropped and the communication network may dynamically re-align itself by adjusting/switching link types between the remaining number of wireless communication nodes in the communication

network to best suit the needs to the wireless communication nodes and the communication network.

[0088] In some embodiments, wireless communication nodes 1120, 1130, 1140, 1150 and 1160 can host multiple modules of the same or different types. For example, one or more of wireless communication nodes 1120, 1130, 1140, 1150 and 1160 can host one Module A and one Module B. Hence, when wireless communication node 1110 makes a point-to-point link using its Module A or Module C with a first communication module (e.g., Module A or C) of wireless communication nodes 1120, 1130, 1140, 1150 and 1160, then a second communication module (e.g., Module B) of wireless communication nodes 1120, 1130, 1140, 1150 and 1160 can simultaneously create point-to-multipoint wireless communication links with other modules of wireless communication nodes in the mesh network that are not shown here. Similarly, when wireless communication node 1110 makes a point-to-multipoint link using its Module B with the first communication module (e.g., Module A or C) of wireless communication nodes 1120, 1130, 1140, 1150 and 1160, then the second communication module (e.g., Module B) of wireless communication nodes 1120, 1130, 1140, 1150 and 1160 can simultaneously create point-to-multipoint wireless communication links with other modules of wireless communication nodes in the mesh network that are not shown here.

[0089] As another example, one or more of wireless communication nodes 1120, 1130, 1140, 1150 and 1160 can host two Module As or Module Cs. Hence, when wireless communication node 1110 makes a point-to-point link using its Module A or Module C with the first Module A or C of wireless communication nodes 1120, 1130, 1140, 1150 and 1160, then the second Module A or Module C of wireless communication nodes 1120, 1130, 1140, 1150 and 1160 can simultaneously create point-to-point wireless communication links with other modules of wireless communication nodes in the mesh network that are not shown here. Similarly, when wireless communication node 1110 makes a point-to-multipoint links using its Module B with the first communication modules (Module A or C) of wireless communication nodes 1120, 1130, 1140, 1150 and 1160, then the second Module A or C of wireless communication nodes 1120, 1130, 1140, 1150 and 1160 can simultaneously create point-to-point wireless communication links with other modules of wireless communication nodes in the mesh network that are not shown here.

[0090] As yet another example, wireless communication nodes 1120, 1130, 1140, 1150 and 1160 can host multiple Module As or Module Cs and a Module B. For instance, one or more of wireless communication nodes 1120, 1130, 1140, 1150 and 1160 can host two Module As or Module Cs and one Module B. Hence, when wireless communication node 1110 makes a point-to-point link using its Module A or Module C with a first Module A or C of wireless communication nodes 1120, 1130, 1140, 1150 and 1160, then a second Module A or Module C of wireless communication nodes 1120, 1130, 1140, 1150 and 1160 can simultaneously create point-to-point wireless communication links with a third communication module (e.g., Module B) of wireless communication nodes 1120, 1130, 1140, 1150 and 1160 can simultaneously create point-to-multipoint wireless communication links with other modules of wireless communication nodes in the mesh network that are not shown here. Similarly, when wireless communication node 1110 makes a

point-to-multipoint link using its Module B with the first communication module (e.g., Module A or C) of wireless communication nodes 1120, 1130, 1140, 1150 and 1160, then the second communication module (e.g., Module A or C) of wireless communication nodes 1120, 1130, 1140, 1150 and 1160 can simultaneously create point-to-point wireless communication links with other modules of wireless communication nodes in the mesh network that are not shown here and a third communication module (e.g., Module B) of wireless communication nodes 1120, 1130, 1140, 1150 and 1160 can simultaneously create point-to-multipoint wireless communication links with other modules of wireless communication nodes in the mesh network that are not shown here.

[0091] It is to be noted that wireless communication links established by Module A or Module C have high reliability due to interference immunity either due to extremely narrow beams or due to transmission of data over ultra-high bandwidth. These features make these links more suitable to carry control information and data for multiple users of a wireless communication mesh network. Hence links established by Module A or Module C can act as a wireless backhaul for a mesh network while links established with Module B can provide access to individual users of a communication network.

[0092] In one embodiment, an entire wireless mesh can be composed of point-to-point links where both links providing backhaul and access have interference immunity. Although such links are more expensive due to the requirement of separate modules to establish individual links, such links are suitable when certain high service quality or reliability is required to be ensured for all customers of the network.

[0093] For example, FIG. 12 shows a location 1200 of a communication network that can be a seed or an anchor home. Location 1200 hosts a wireless communication node 1201 that contains a total of 6 communication modules that belong to either Module A or Module C. Hence wireless communication node 1201 is capable of establishing six point-to-point links. As shown, wireless communication node 1201 uses a 1st and 4th Module A/Module C to establish connections with location 1210 and location 1260 that serve as backhaul links, while wireless communication node 1201 uses a 2nd, 3rd, 5th and 6th Module A/Module C to establish point-to-point links with location 1220, 1230, 1250 and 1240 to provide access links. In this respect, links between locations 1200 and 1220, locations 1200 and 1230, locations 1200 and 1240, and locations 1200 and 1250 only carry data for individual users, whereas links between locations 1200 and 1260 and locations 1200 and 1210 carry signaling and data for all the locations including 1200, 1210, 1220, 1230, 1240, 1250 and 1260.

[0094] In another embodiment, an entire wireless mesh can be composed of combination of point-to-point links and point-to-multipoint links where point-to-point links act as backhaul links and point-to-multipoint links act as access links to individual users. Although such wireless mesh networks due to presence of point-to-multipoint links provide interference immunity to all the users of the communication network, such wireless mesh networks are less expensive due to the non-requirement of separate modules to establish individual links.

[0095] For example, FIG. 13 shows a location 1300 of a communication network that can be a seed or an anchor home. Location 1300 hosts a wireless communication node

1301 that contains a total of 4 communication modules that belong to either Module A/Module C or Module B. Hence this wireless communication node is capable of establishing two point-to-point links and two point-to-multipoint links. As shown, wireless communication node **1301** uses a 1st and 4th Module A/Module C to establish connections with location **1310** and location **1360** that serve as backhaul links, while wireless communication node **1301** uses a 2nd Module B to establish point-to-multipoint links with locations **1320**, **1330** and uses a 3rd Module B to establish point-to-multipoint links with locations **1350** and **1340** to provide access links. In other words, links between locations **1300** and **1320**, locations **1300** and **1330**, locations **1300** and **1340** and locations **1300** and **1350** only carry data for individual users, whereas links between locations **1300** and **1360** and locations **1300** and **1310** carry signaling and data for all the locations including **1300**, **1310**, **1320**, **1330**, **1340**, **1350** and **1360**.

[0096] Referring to FIG. 14, another possible embodiment of wireless communication node **131** of FIG. 1 is shown as wireless communication node **1400**. Wireless communication node **1400** comprises a single module labeled as “Module D.” Module D comprises base band unit or digital unit **1401** which runs the physical layer level protocol including digital modulation/demodulation (modem) and other higher layer protocols such as MAC layer, etc. Base band unit **1401** interacts with other nodes of the communication network that are external to the wireless communication node **1400** via wired medium.

[0097] Module D also includes RF unit **1402**, which among other things processes intermediate frequency (IF) signals and defines the frequency range of the radio signals that can be transmitted or received with the Module D. RF unit **1402** is capable of operating over a wide range of frequencies (e.g., 5 Ghz band frequencies ranging from 5 Ghz to 6 Ghz).

[0098] Further, as shown, Module D also comprises antenna unit **1403** which performs the transmission and reception of over the air radio signals. Antenna unit **1403** is capable of transmitting and receiving extremely narrow beam of signals. Antenna unit **1403** may be constructed with either 1-dimensional or 2-dimensional antenna element arrays that have excellent properties of controlling the directionality of radio signals using beam forming and can propagate even in a non-line of sight environment. Module D with the help of antenna unit **1403** is capable of establishing point-to-multipoint links with a tower capable of performing massive MIMO (multiple input multiple output) beams. In one embodiment, Module D can be designed and manufactured at least in part using ASIC (Application specific integrated circuit) and an integrated RF unit called RFIC.

[0099] Referring to FIG. 15, an example of multiple Module Ds connected to a tower **1500** is shown. Specifically, wireless communication node **1501** hosting a Module D described above is connected to tower **1500** via massive MIMO beam link **1510** that can be both line-of-sight and non-line-of-sight, wireless communication node **1502** hosting a Module D described above is connected to tower **1500** via massive MIMO beam link **1520** that can be both line-of-sight and non-line-of-sight, and wireless communication node **1503** hosting a Module D described above is connected to tower **1500** via massive MIMO beam link **1530** that can be both line-of-sight and non-line-of-sight. The tower **1500**

is equipped with a Massive MIMO module that can create multiple bi-directional narrow beam links simultaneously in all directions with **360** degrees of coverage area. In one embodiment, tower **1500** can operate in the 5 Ghz band including frequencies ranging from 5000 Mhz to 6000 Mhz. In other embodiments, tower **1500** and associated wireless communication nodes **1501**, **1502** and **1503** can operate within a different frequency band.

[0100] It should be understood that while FIG. 15 shows only one tower and three wireless communication nodes hosting Module D in the network, a given network can have multiple towers similar to tower **1500** and these towers can each be connected to a large number of wireless communication nodes hosting various other modules.

[0101] In accordance with the present disclosure, the route that a particular packet takes from a source to a destination may be dynamically selected based on factors including but not limited to link quality, loading, latency etc. For example, referring to FIG. 16, communication system **1600** is shown that is similar to communication system **100** and has all the components described in the context of FIG. 1. Additionally, system **1600** of FIG. 16 includes a tower **1610** which is similar to tower **1500** described in the context of FIG. 15. In contrast to communication system **100** in FIG. 1, wireless communication nodes **131**, **132**, **133**, **134** and **135** host an additional Module D besides Module A/Module B or Module C that enables these wireless communication nodes to optionally establish bi-directional links with features described in the context of FIGS. 14-15 with tower **1610** using massive MIMO beamforming capabilities. Such links labeled as **1601**, **1602**, **1603**, **1604** and **1605** can work in both line-of-sight and non-line of sight environment and can provide alternate communication path to wireless communication nodes **131**, **132**, **133**, **134** and **135** in an event where point-to-point or point-to-multipoint links that connect one wireless communication node to a peer wireless communication node to form a mesh network fails or experience performance degradation due to various reasons including but not limited to change on the line-of-sight profile of millimeter wave link between two wireless communication nodes.

[0102] In FIG. 16, only one tower (i.e., tower **1610**) capable of massive MIMO point-to-multipoint communication is shown to be connected to five wireless communication nodes **131-135**. However, it should be understood that a communication system can also have more than one tower, each connected to multiple different wireless communication nodes hosting various other modules.

[0103] In areas within tower **1500**'s (and other towers of same type) coverage area, a given communication network can initially start in a point-to-multipoint manner, where tower **1500** (and other towers of same type) provides access to individual customers using sub 6 Ghz massive MIMO point-multipoint beams. Later, nodes in the given communication network can opportunistically connect with other nodes using regular modules (e.g. Module A/Module B/Module C) in the presence of line-of-sight. This way, the given communication network may evolve into a mesh network with point-to-point and point-to-multipoint connection between nodes in addition to each communication node having a path directly (non-line-of-sight) to tower **1500** (and other towers of same type) that fall within the coverage area.

[0104] One of ordinary skill in the art will appreciate that a route a given packet takes from a source to a destination

in this network may be optimized by considering various factors including latency, congestion, reliability etc. One of ordinary skill in the art will also appreciate that a given communication network can later add seed homes (e.g., seed homes **111** and **115** in FIG. 1) along with tower/fiber access points **101** and **102** to provide alternate backhaul as per need basis.

[0105] In another embodiment, instead of providing massive MIMO point-to-multipoint networking capability using a terrestrial tower, point-to-multipoint massive MIMO capability to networks wireless communication nodes can also be provided by a satellite for example a low earth orbit (LEO) satellite. For example, referring to FIG. 17, communication system **1700** is shown that is similar to communication system **100** and has all the components described in the context of FIG. 1. Additionally, system **1700** of FIG. 17 includes a satellite **1710** which is capable of massive MIMO transmission and reception on frequencies including but not limited to 5-6 Ghz, similar to tower **1500** described in the context of FIG. 15. In contrast to communication system **100** in FIG. 1, wireless communication nodes **131**, **132**, **133**, **134** and **135** host an additional Module D (besides Module A/Module B or Module C) that enables these wireless communication nodes to optionally establish bi-directional links with features described in the context of FIGS. 14-15 with satellite **1710** using massive MIMO beamforming capabilities. Such links labelled as **1701**, **1702**, **1703**, **1704** and **1705** can provide alternate communication path to wireless communication nodes **131**, **132**, **133**, **134** and **135** in an event where point-to-point or point-to-multipoint links that connect one wireless communication node to a peer wireless communication node to form a mesh network fails or experience performance degradation due to various reasons including but not limited to change on the line-of-sight profile of millimeter wave link between two wireless communication nodes.

[0106] In FIG. 17, only one satellite **1710** capable of massive MIMO point-to-multipoint communication is shown to be connected to five wireless communication nodes **131-135**. However, it should be understood that a communication system can also have more than one satellite, each connected to multiple different wireless communication nodes hosting various other modules.

[0107] In another embodiment, some of the wireless communication nodes that provide backhaul functionality can be equipped with multiple communication modules that enable these wireless communication nodes to provide transport backhaul data between an end user and a network using multiple different types of communication links. For example, referring to FIG. 18, communication system **1800** is shown that is similar to communication system **100** and has all the components described in the context of FIG. 1. Additionally, system **1800** of FIG. 18 includes a satellite **1810** which is capable of massive MIMO transmission and reception on frequencies including but not limited to 5-6 Ghz, similar to tower **1500** described in the context of FIG. 15. System **1800** also includes a massive MIMO cable tower **1820** which is also similar to tower **1500** described in the context of FIG. 15.

[0108] In contrast to communication system **100** in FIG. 1, wireless communication nodes **131**, **132**, **133**, **134** and **135** host an additional Module D (besides Module A/Module B or Module C) that enables these wireless communication nodes to optionally establish bi-directional links with fea-

tures described in the context of FIGS. 14-15 with satellite **1810** and tower **1820** using massive MIMO beamforming capabilities. Such links labeled as **1801**, **1802**, **1803** and **1804** can provide an alternate communication path to wireless communication nodes **131**, **132**, **133**, **134** and **135** in an event where point-to-point or point-to-multipoint links that connect one wireless communication node to a peer wireless communication node to form a mesh network fails or experience performance degradation due to various reasons, including but not limited to change in the line-of-sight profile of a millimeter wave link between two wireless communication nodes. Specifically, satellite **1810** in FIG. 18 is connected to seed home **115** using wireless communication node **135** via link **1804** and to anchor home **112** using wireless communication node **132** via link **1803**. Seed home **115** thus has multiple options to route backhaul traffic to the network.

[0109] In one embodiment, seed home **115** using wireless communication node **135** at a given time can pick a satellite link **1804** to transport backhaul data, and based on some trigger at a different time, instruct wireless communication module **135** to switch links for backhaul data transmission from **1804** to a point-to-point or point-to-multipoint millimeter wave (e.g. E-band) based link coupled to tower/fiber access point **102**. Such trigger may include latency, bandwidth, packet loss requirements, etc. of a particular application.

[0110] FIG. 18 also shows an end user home **113** where wireless communication node **133** transports the data using anchor home **112**'s wireless communication node **132**. Wireless communication node **132** is shown to have multiple options to transport end user data of home **113**, including direct satellite link connection using **1803**, in-direct satellite link connection using **1804** via anchor node **135**, or though point to point/multi-point connections using millimeter wave through towers **101** or **102** via seed homes **111** and **115** respectively, among other options.

[0111] In one embodiment, wireless communication node **132** can dynamically switch its connection link to route data to and from end user home **113**. For example, due to some trigger similar to the triggers described above, wireless communication node **132** can dynamically switch from satellite link **1803** to satellite link **1804** via wireless communication node **135** to transport data to and from end user home **113**.

[0112] It should be understood that links **1803** and **1804** can be part of same massive MIMO beam or links **1803** and **1804** can be part of different massive MIMO beams. It should also be understood that satellite links **1802** and **1804** can use the same frequency range of communications or can operate in different frequency ranges. Further, while FIG. 18 shows only one satellite (i.e., satellite **1810**) capable of massive MIMO point-to-multipoint communication that is connected to two wireless communication nodes **132** and **135**, it should be understood that a communication system can also have more than one satellite, each connected to multiple different wireless communication nodes hosting various other modules.

[0113] As further shown in FIG. 18, tower **1820** is connected to seed home **111** using wireless communication node **135** via link **1804** and to anchor home **112** using wireless communication node **131** via link **1801** and to anchor home **114** using wireless communication node **134** via link **1802**. This provides anchor home **114** with options to route packets

to the network in multiple ways including (a) through point-to-point or point-to-multipoint millimeter wave-based links **152** or **155**, and (b) via direct massive MIMO based link to tower **1820** via link **1802**.

[0114] Similarly, seed home **111** has multiple options to route backhaul traffic to the network. In one embodiment, seed home **111** using wireless communication node **131** at a given time can pick a satellite link **1801** to transport backhaul data and based on some trigger at a different time, instruct wireless communication module **131** to switch links for backhaul data transmission from **1801** to a point-to-point or point-to-multipoint millimeter wave (e.g. E-band) based link coupled to tower/fiber access point **101**. Such trigger may include latency, bandwidth, packet loss requirements, etc. of a particular application.

[0115] In FIG. **18**, only one tower (i.e., tower **1820**) capable of massive MIMO point-to-multipoint communication is shown to be connected to two wireless communication nodes **131** and **134**. However, it should be understood that a communication system can also have a different number of massive MIMO towers, each connected to multiple different wireless communication nodes hosting various other modules.

[0116] In another embodiment, one or more wireless communication nodes described above and discussed with respect to FIGS. **1-18** may additionally be an edge computing node by hosting a processor (separate or shared), memory, digital contents, software, and storage, among other components for computing, and other required operations for edge computing, in addition to the high speed low latency networking capability that has already been described above. This enables a given communication system to provide cloud services in a distributed manner closer to an end user as wireless communication nodes are distributed across the network and provide an interface between the network and an end-user. This memory unit can store a copy of local digital contents and can additionally store portions of digital content that are not local. The non-local digital contents among other things can include digital content belonging to other nodes. This provides content redundancy in a communication system. Hence, when an end user of a communication system requests for digital content, then this edge computing mechanism allows a request to be fulfilled in a variety of different ways, including a request processed by a local node and/or remote node based on various criteria including but not limited to latency, network congestion, etc. of the application making the request.

[0117] In another embodiment, one or more wireless communication nodes described above and discussed with respect to FIGS. **1-18** can additionally be a blockchain node by hosting a computer comprising at least one processor, memory, digital content, software, etc., which is connected to a blockchain network comprising a client that is capable of storing, validating and/or relaying transactions in addition to the high-speed low latency networking capability that has already been described above. This enables the communication system and its nodes described in this disclosure and discussed in the context of FIGS. **1-18** to provide an ideal platform for blockchain databases, enterprise blockchain databases, permissioned/private blockchains, hybrid and other similar types of databases given that (1) file/data/record storage space is inherently distributed as wireless communication nodes are distributed across the geographi-

cal coverage area and (2) low latency communication between the nodes and across the network due to high speed wireless links enable improved latency and improves the transaction throughput of the blockchain based databases.

[0118] In another preferred embodiment, one or more wireless communication nodes can additionally act as blockchain-based distributed data storage node by adding dedicated or shared storage capacity capability to these nodes. One key advantage of implementing blockchain-based distributed data storage on a given communication system and the wireless communication nodes described in this disclosure is that storage nodes are inherently distributed, and due to the low latency and high bandwidth of the wireless communication links between the wireless communication node described above and the proximity of the storage location nodes to an end-user, accessing the data content can be faster compared to other approaches.

[0119] In accordance with the present disclosure, the wireless mesh network node equipment (point-to-point link modules, point-to-multipoint link modules, multiple point-to-point link modules, combination of multiple point-to-point and point-to-multipoint links, antennas for cellular small cells/CPEs and mmWave equipment, cable, mounts, power supply boxes, etc.) that gets deployed and installed on a rooftop of a private infrastructure such as a single-family home can be consumer financed. For instance, in case of a customer meeting a certain credit score threshold (or any other credit checking criteria), the equipment required to add a millimeter wave mesh node at the customer's premises (i.e., to add the customer to the wireless mesh network) and provide high speed internet service may be financed by a bank on the behalf of the customer, and the customer may agree with the financing bank to re-pay the amount financed by the bank over a certain time period by making periodic (e.g. monthly) payments based on the terms and conditions of the agreement. This way, the customer becomes owner of the equipment (a wireless mesh network node) once the full financed amount is made to the financing bank. This customer can in one embodiment lease back the wireless mesh network node equipment installed on its property to the wireless mesh network operator that installed the wireless mesh network equipment on its property and provide high speed internet data service. In another embodiment, this customer can lease back the wireless mesh network node equipment installed on its property to the wireless mesh network operator that installed the wireless mesh network equipment on its property and provide high speed internet data service for a certain term (e.g., 18 months, 24 months, 36 months, etc.).

[0120] In some instances, this customer may be required to lease back the equipment to only that operator which originally installed the equipment at the customer location and provided high speed internet data services. In other instances, this customer can lease back the equipment to any wireless internet network operator. In another instance, lease back of the equipment to an operator other than the operator which originally installed the network equipment at the customer location may only occur with the permission of the wireless internet network operator that originally installed that equipment at customer location. In yet another instance, such lease back to a different wireless internet network operator may only occur after expiration of the lease term with the original wireless internet network operator.

[0121] For a wireless internet network operator building and operating a wireless mesh network, the type of customer financing-based network deployment described above becomes a crowd sourcing or crowdfunding-based infrastructure roll out mechanism, where instead of one or few large entities, CAPEX is sourced from a pool of individuals who in some instances are the customers of the wireless mesh network operator. Such customers can get high speed internet data service from the wireless mesh network operator (operating using ptp/ptmp modules, other communication nodes and equipment and various variations discussed earlier in this disclosure) at a subsidized/discounted rate. In certain cases, such customers may get two separate bills periodically, one for the high-speed internet data service and other for the equipment financing from bank. In another case, customers can get a single consolidated bill from a wireless mesh operator.

[0122] In some instances, all customers of a wireless mesh operator can be based on consumer financing explained above in a neighborhood or market where wireless mesh operator offers its high-speed internet data service. In other instances, wireless mesh network's customers in a market or neighborhood can be financed through a variety of different ways including operator financing where wireless mesh operator pays for the equipment of the wireless mesh node, financed through bundling with a private utility or service that has a relatively smaller market size (e.g. home security/automation, solar energy, etc.) compared to market size of the high speed internet where a bundled service is offered and wireless mesh operator uses the marketing/sales commission received from the private utility or service provider to fund the wireless mesh node equipment, financed through the revenue generated from running blockchain platform based services on the wireless mesh network nodes along with the consumer/customer based financing that is explained earlier in the disclosure.

[0123] Further, in accordance with the present disclosure, the communications equipment including various types of ptp/ptmp modules, cellular small cell, etc. that were described above can be used to establish multiple point-to-point and/or point-to-multiple links where both network nodes of a wireless link, one from where a link originates and the second from where a link terminates (in general, nodes can switch roles dynamically between link originator and link terminator based on the direction of data flow), are located at the different customer locations and providing high speed internet service to the dwellers of the property where wireless mesh network node is deployed and installed. In some cases, one of the two nodes of the link can be at a location where the deployed equipment provides high speed internet service to the dwellers of the property at that location. In other instances, both nodes of the link may be at a location where the deployed equipment does not provide high speed internet service to the dwellers of the property at that location.

[0124] It should be understood that the length of the communication links of a wireless mesh network describe above may vary. For instance, the length of the communication links of a wireless mesh network established with the help of the various communication modules and equipment described above may be less than 300 meters on average. Alternatively, the length of the communication links of a wireless mesh network can be greater than 300 meters on average as well.

[0125] In accordance with the present disclosure, further disclosed herein are communication modules that employ direct RF (microwave/millimeter wave)-to-optical and direct Optical-to-RF (microwave/millimeter wave) conversion. In one example implementation, the high-speed photo detectors can be used that directly translate an optical signal into a microwave signal. One of ordinary skill in the art will appreciate that other approaches can be used for direct optical-to-RF conversion. Similarly, a dipole antenna directly coupled to a plasmonic modulator allows direct conversion from the RF to the optical world. One of ordinary skill in the art will appreciate that different approaches can be used for direct conversion of RF signals to optical signals. This direct optical-to-RF and direct RF-to-Optical conversion modules eliminate the need of the use of analog to digital and digital to analog (ADC/DAC) modules that are required by traditional modem implementations. These mixed signal components (i.e., ADC/DAC) consume high amount of power and also increase the cost as each antenna is required to be connected to a separate ADC/DAC modules.

[0126] FIG. 19 shows a communication module based on direct RF-to-Optical and direct Optical-to-RF conversion. Communication module of FIG. 19 contains a single direct RF-to-Optical sub-module and a single Optical-to-RF sub-module. However, communication module of FIG. 19 can host any integer number of direct RF-to-Optical sub-modules greater than or equal to zero and any integer number of direct Optical-to-RF sub-modules greater than or equal to zero. In one example embodiment, this direct RF-to-Optical and direct Optical-to-RF conversion technology can be implemented is an integrated Circuit (IC) or chip.

[0127] Based on the above explanation with respect to the example communication module of FIG. 19, in an example embodiment, the core of the wireless mesh network can be a wireline optical or wired router/switch where each port is mapped, either through a direct connection or over optical/wired line, to an individual direct conversion Optical-to-RF or RF-to-Optical chip that then focuses, on both receiver and transmitter side, all RF energy into a high gain narrow beam that can be both fixed or steerable. In one example embodiment, a standard 8-port×10G router/switch could be used, with one port being used as a data drop to local building/site and the other 7 ports being connected over a fiber optic cable to various Optical-to-RF or RF-to-Optical end points that are located at multiple distributed locations external (and/or internal) on/in the building/site as shown in FIG. 20. One of ordinary skill in the art will understand that the router/switch can have a different number of ports as well.

[0128] These multiple distributed locations can be determined in advance based on the use of connection potentiality optimization algorithms, where the algorithm understands the relationship between end point placement and potentially connection partners. Also, the individual PtP beams can be dynamically steered among potential PtP connection partners to facilitate path optimization algorithms and/or to respond to network congestion and/or network element failures. In one embodiment, these Optical-to-RF or RF-to-Optical end points that establish ptp/ptmp beams can be placed below a roof's eaves and in other embodiments, these end points can be placed above a roof's eaves. In some other embodiments, some of the Optical-to-RF or RF-to-Optical end points can be placed below a roof's eaves and some can

be placed above a roof's eaves and actual placement may depend upon the line-of-sight profile of the location/site.

[0129] It should be understood that the example communication module discussed in the context of FIGS. 19-20 can be implemented in other communication modules that were discussed in the context of FIGS. 1-18. For instance, the modules discussed in the context of FIGS. 1-18 can have direct RF-to-Optical and direct Optical-to-RF technology embedded such that the narrow beam, extremely narrow beam, and/or ptp/ptmp/multiple ptp links can be established without the need for ADC/DAC mixed signal circuitry that consumes a high amount of power and requires to be connected individually with each antenna.

[0130] In accordance with the present disclosure, a modified version of the communication nodes discussed earlier for building a wireless mesh network will now be discussed. In one embodiment, a communication node can be a flexible millimeter wave radio equipment capable of establishing multiple point-to-point and/or point-to-multipoint links operating over millimeter wave frequencies and can comprise 3 different sub-modules: (1) digital/network module, (2) point-to-point radio module, and (3) point-to-multipoint radio module. A digital/network module is responsible for interfacing the above millimeter wave radio box (communication node) with a backhaul or fiber network. Specifically, it provides switching capability to direct traffic between the point-to-point or point-to-multipoint radio modules of the millimeter wave radio box (communication node) and the fiber or backhaul network. The connectivity between a single or multiple point-to-point and/or point-to-multipoint radio modules of the millimeter wave radio box and the backhaul or fiber network can be based over a variety of interfaces including but not limited to PCI/PCIe express bus interface and ethernet.

[0131] In one embodiment, PCI/PCIe can be used when a point-to-point or point-to-multi-point radio that needs to be connected is enclosed in the same box with a digital/network module and separation between the digital/network module and the point-to-point module is limited to few inches such as 3-6 inches or less.

[0132] In one embodiment, a digital/network module provides connectivity to a single point-to-point or point-to-multipoint module over a single PCI/PCIe bus interface. In a different embodiment, a digital/network module provides connectivity to 3 point-to-point or 3 point-to-multipoint or a combination of 3 point-to-point/point-to-multipoint modules over three separate PCI/PCIe bus interfaces. In another embodiment, a digital/network module provides connectivity to N point-to-point or N point-to-multipoint or a combination of N point-to-point/point-to-multipoint modules over N separate PCI/PCIe bus interfaces, where N is a positive integer number greater than zero.

[0133] An ethernet interface such as an RJ45 port with multi-gigabit support, including but not limited to 1 Gb, 2.5 Gb, 5 Gb, 10 Gb, etc., can be used to connect point-to-point or point-to-multipoint radio modules with a digital/network module. In one embodiment, an ethernet interface can be used when the point-to-point or point-to-multi-point radio that needs to be connected is not enclosed in the same box with a digital/network module and separation between digital/network module and the point-to-point module is greater than 3-6 inches. In some embodiments, the length can be 10 meters or more.

[0134] In one embodiment, a digital/network module provides capability of connecting up to 4 point-to-point/point-to-multipoint radios or up to 3 point-to-point/point-to-multipoint radio and a small cell over 4 ethernet interfaces. In a different embodiment, a digital/network module provides capability of connecting up to N point-to-point/point-to-multipoint radios or up to N-1 point-to-point/point-to-multipoint radio and a small cell over N ethernet interfaces, where N is a positive integer number greater than zero. Digital/network module also contains SFP/SFP+ interface or any other interface to connect digital/network module with fiber/backhaul network.

[0135] The point-to-multipoint radio module of the communication node discussed above is responsible for establishing point-to-multipoint millimeter wave based bi-directional links to connect to peer millimeter wave radios in the network. Point-to-multipoint radio module comprises a baseband sub-module and RF module. Baseband module handles the baseband processing and among other aspects is responsible for baseband processing related to beamforming. RF module contains phased antenna array that works in conjunction with baseband module to generate point-to-multipoint millimeter wave beams.

[0136] The point-to-point radio module of the communication node described above is responsible for establishing point-to-point millimeter wave based bi-directional links to connect to a peer millimeter wave radio in the network. The point-to-point radio module comprises a baseband sub-module, RF module and beam narrowing module. The baseband module handles the baseband processing and, among other aspects, is responsible for baseband processing related to beamforming. RF module contains phased antenna array that works in conjunction with baseband module to generate point-to-point millimeter wave beam. A beam narrowing module is responsible for narrowing the beam by focusing most of the radiated signal energy in the desired direction and lowering the antenna side lobes to minimize the interference in a mesh network.

[0137] In one embodiment, the beam narrowing module can be a lens antenna integrated with an RF module. In another embodiment, the beam narrowing module can be a parabolic antenna integrated with an RF module. In yet another embodiment, the beam narrowing module could be a module other than a lens or parabolic antenna and rely on a different approach to narrow the beam originating from a phased array based RF module.

[0138] Referring to FIG. 21, a logical block diagram of the communication module described above is shown. As explained earlier, a flexible millimeter wave radio node contains within an enclosure (typically outdoor) a digital/network module that has a network processing unit and is configured to provide network switch operations between the fiber optic backhaul interface and the point-to-point or point-to-multipoint radio modules either connected via PCI/PCIe interface or via multi gigabit ethernet ports. A flexible millimeter wave radio module also contains within the enclosure 3 point-to-point or point-to-multipoint radios. For providing mesh network deployment flexibility, a node can also be connected to external ptp/ptmp radios via ethernet ports. A node can be solar powered or can be powered via electric power outlet of the home where the node is installed. FIG. 21 also shows that this flexible millimeter wave radio node may only need a single network processing unit (NPU) that controls all the point-to-point or point-to-multipoint RF

modules either connected via a PCI/PCIe interface or via a multi gigabit ethernet interface. Hence this example flexible millimeter wave radio node removes the need for using a dedicated NPU for each ptp/ptmp RF module.

[0139] FIG. 22 shows a block diagram of a ptmp radio module of the communication node described above. As shown, this radio module contains a baseband module and a RF module that has the phase antenna array for providing beamforming capability.

[0140] FIG. 23 shows a block diagram of the ptp radio module of the communication node discussed above. This radio module contains a baseband module, an RF module that has the phase antenna array for providing beamforming capability, along with a beam narrowing module. The beam narrowing module, based on various techniques discussed earlier, narrows the beam generated by the phase antenna array of the RF module.

[0141] Referring to FIG. 24, various different use cases of the communication node described above and explained in the context of FIGS. 21-23 is shown. FIG. 24 shows a mesh network comprising 5 communication nodes 3700. Communication nodes 3700 may each be a flexible millimeter wave communication node that has been discussed earlier.

[0142] At “Site A” of the mesh network, a communication node 3700 may be solar powered and mounted on the pole. This node 3700 at Site A may have 3 ptp links generated by 3 ptp radio modules integrated with the digital/network module. At “Site B,” a communication node 3700 may be powered with an electric power outlet of the home and may have one ptp link via a single integrated ptp radio module and 2 ptmp links via two ptmp radio modules that are not integrated with a digital/network module but instead connected via ethernet interface to the communication node. Similarly, at “Site C,” a communication node 3700 may be powered with an electric power outlet of the home and may have two ptp links via two integrated ptp radio module and one ptmp radio module integrated with a digital/network module. At “Site E,” a communication node 3700 may be powered with an electric power outlet of the home and may have two ptp links via two integrated ptp radio module. Further, at “Site D,” a communication node 3700 may be powered with an electric power outlet of the home and may have two ptp links via two integrated ptp radio module and one ptmp radio module integrated with the digital/network module.

[0143] Referring to FIG. 25A, another use case of the communication node described above is shown. In particular, FIG. 25A shows an example mesh network that includes communication nodes 3700 at the 5 sites previously described with respect to FIG. 24, as well as an additional communication node 3700 at “Site A2.” Similar to communication node 3700 at “Site A,” communication node 3700 at “Site A2” may be mounted on a pole (among other possibilities).

[0144] Based on the preceding disclosure (e.g., the disclosure in connection with FIGS. 5-7, 9-11, 13, and 16-18), one of ordinary skill in the art will appreciate that each communication node 3700 at a given site may have the capability to communicate with multiple other communication nodes at multiple other sites. For instance, communication node 3700 at “Site B” may have the capability to communicate with the respective communication nodes 3700 at both “Site A” and communication node 3700 at “Site A2.” Similarly, the respective communication node 3700 at

each of “Site C,” “Site D,” and “Site E” may have the capability to communicate with the respective communication nodes 3700 at both of “Site A” and “Site A2.”

[0145] Furthermore, based on the preceding disclosure (e.g., the disclosure in connection with FIGS. 5, 11, and 18), one of ordinary skill in the art will appreciate that each communication node 3700 at a given site (e.g., communication node 3700 at “Site B”) may have the capability to dynamically switch its active communication link from a first communication node 3700 at a first site (e.g., communication node 3700 at “Site A”) to a second communication node 3700 at a second site (e.g., communication node 3700 at “Site A2”) based on some trigger that is similar to the triggers described above (e.g., changes in link condition such as a change from a LOS path to a non-LOS path due to a change in environment, increased interference, instructions from higher layers, latency, bandwidth, and/or packet loss requirements of a particular application, etc.).

[0146] For instance, in the scenario shown in FIG. 25A, the respective communication node 3700 at each of “Site B,” “Site C,” “Site D,” and “Site E” may initially be configured to actively communicate with the communication node 3700 at “Site A” (which may function to route backhaul traffic to and/or from such other sites). However, at some later point in time, the communication node 3700 may dynamically switch its active communication link from the communication node 3700 at “Site A” to the communication node 3700 at “Site A2” (which may also function to route backhaul traffic to and/or from such other sites) due to some trigger similar to the triggers described above. Such a scenario is shown in FIG. 25B.

[0147] It should be understood that FIGS. 24-25 are described in such a manner for the sake of clarity and explanation and that the example mesh networks described in FIGS. 24-25 may take various other forms as well. For instance, the example mesh networks may include more or less communication nodes, and a given communication node may take various other forms and may be mounted in various other manners and/or mounted on various other objects as well (e.g., mounted on a pedestal). Further, in line with the preceding disclosure, one or more of the communication nodes (e.g., the communication nodes 3700 at “Site A” and “Site A2”) may be mounted to an object that is at or near a fiber access point. Further yet, the example mesh networks may have various different configurations of ptp or ptmp modules either integrated or connected via an ethernet interface and powered via various different power options.

[0148] Another important aspect of communication node 3700 is that the integrated radio modules can be pluggable. In other words, based on a specific use case, the number and types of radio modules integrated with a digital/network module via PCI/PCIe interface can easily be changed by plugging in the desired number and type of radio modules with full flexibility instead of having one specific configuration.

[0149] So far the modified version of communication nodes discussed above and also described in the context of FIGS. 21-25 assumes that the point-to-point or point-to-multipoint modules connected to a digital/network module with an NPU via a high speed interface (e.g., PCI/PCIe/Thunderbolt) are also located inside a same enclosure. It should be understood that the point-to-point or point-to-multipoint modules connected to a digital/network module via high speed interface can also be located outside the

digital/network module with the NPU and inside an independent box/enclosure connected via an outdoor cable supporting the PCI/PCIe/Thunderbolt high speed communication protocol to the enclosure of the digital/network module.

[0150] As one example, FIG. 26 depicts a modified version of a flexible millimeter wave radio box, where the point-to-point or point-to-multipoint RF modules are located outside a digital/network module with NPU enclosure and inside separate independent box/enclosure and connected via an outdoor wired cable capable of supporting high speed communication interface (e.g., PCI/PCIe/Thunderbolt Interface). As shown, 3 point-to-point or point-to-multipoint modules are connected via PCIe/Thunderbolt interfaces to the digital/network module with the NPU using a compatible outdoor cable.

[0151] In general, it should be understood that N number of point-to-point or point-to-multipoint modules in separate independent enclosures can be connected via a PCIe/Thunderbolt compatible outdoor cable, where N is an integer greater than zero. It should also be understood that the length of the outdoor cable compatible with high speed communication protocol, such as PCIe/thunderbolt, depends on the maximum limit defined by the technology. In one embodiment, PCIe/thunderbolt cable can be up to 3 meters. In other embodiments, the length of the outdoor PCI/PCIe/thunderbolt compatible cable can be less than or greater than 3 meters.

[0152] In yet another embodiment of the present disclosure, a wireless mesh network may include ultra-high-capacity nodes that are capable of establishing ultra-high-capacity links (e.g., point-to-point or point-to-multipoint bi-directional communication links) using a millimeter wave spectrum, including but not limited to 28 Ghz, 39 Ghz, 37/42 Ghz, 60 Ghz (including V band), or E-band frequencies, as examples. These ultra-high-capacity links may have a larger range as compared to other point-to-point or point-to-multipoint links, including but not limited to point-to-point or point-to-multipoint links of the type discussed above with reference to FIGS. 1-26.

[0153] For instance, as one possibility, a point-to-point or point-to-multipoint link of the type discussed above with reference to FIGS. 1-26 may have an average range of up to 100 meters, whereas an ultra-high-capacity link may have a range of more than 100 meters. As another possibility, a point-to-point or point-to-multipoint link of the type discussed above with reference to FIGS. 1-26 may have an average range of up to 500 meters, whereas an ultra-high-capacity link may have a range of more than 500 meters. As yet another possibility, a point-to-point or point-to-multipoint link of the type discussed above with reference to FIGS. 1-26 may have an average range of up to 1000 meters, whereas an ultra-high-capacity link may have a range of more than 1000 meters.

[0154] However, in other implementations, it is possible that the length of an ultra-high-capacity link may be similar to the length of a point-to-point or point-to-multipoint links of the type discussed above with reference to FIGS. 1-26, but may nevertheless provide higher capacity such that a fewer number of ultra-high-capacity nodes/links may be used (as compared to the point-to-point or point-to-multipoint nodes/links of the type discussed above with reference to FIGS. 1-26) to build a main high capacity backbone through the mesh (i.e., the ultra-high-capacity nodes/links may be more sparse).

[0155] The higher capacity and/or extended range of these ultra-high-capacity nodes/links may be achieved via various advanced signal processing techniques, including but not limited to multiple input multiple output (MIMO) such as 2x2 MIMO, 4x4 MIMO, 8x8 MIMO or an even higher order MIMO, use of vertical and horizontal polarization (V & H), higher switch capacity of the digital network module due to higher processing power such as support of 8x25 Gbps port (200 Gbps aggregate traffic flow), higher order modulation including 16 QAM, 64 QAM, 256 QAM, 512 QAM, 1024 QAM, orbital angular momentum (OAM) multiplexing, and/or higher antenna gains, among other possibilities. Further, in some implementations, the higher capacity and/or extended range of these ultra-high-capacity nodes/links can be achieved using a subset of the advanced signal processing techniques mentioned above.

[0156] These ultra-high-capacity nodes/links may be used in conjunction with other point-to-point and/or point-to-multipoint links, including but not limited to point-to-point or point-to-multipoint links of the type discussed above with reference to FIGS. 1-26, to design a multi-layer wireless mesh network.

[0157] To illustrate with an example, FIG. 27 shows one example of a multi-layer wireless mesh network in which triple-compound links represent the ultra-high-capacity links described above, double-compound rings represent point-to-point links of the type discussed above with reference to FIGS. 1-26, and single-line links represent point-to-multipoint links of the type discussed above with reference to FIGS. 1-26.

[0158] As shown in FIG. 27, longer ultra-high-capacity links may be used bring a high level of capacity to the wireless mesh network, which can then be delivered to an end user/customer via a shorter point-to-point or point to multi point link (which may not be ultra-high-capacity). It should also be understood that while the point-to-multipoint links may primarily serve to provide flexibility in building the wireless mesh network due to the capability of beam steering and ability to establish multiple links from a single radio, these point-to-multipoint links may also be used to indirectly connect two point-to-point links via multiple point-to-multipoint link hops that can add additional reliability to the network.

[0159] Further, it should be understood that a multi-layer wireless mesh network such as the one illustrated in FIG. 27 can be deployed in various manners. For instance, in one implementation, different layers of the multi-layer mesh network can be deployed in parallel. In another implementation, different layers of the multi-layer wireless mesh network can be deployed in different phases. For example, a deployment approach for a multi-layer wireless mesh network may involve first building a core network backbone (e.g., an ultra-high-speed network) using ultra-high-capacity nodes/links and then densifying the network during one or more subsequent phases using other types of point-to-point or point-to-multipoint nodes/links, including but not limited to point-to-point or point-to-multipoint radio links of the type discussed above with reference to FIGS. 1-26. In another example, a deployment approach for a multi-layer wireless mesh network may involve first building a network of point-to-point nodes/links that are not ultra-high capacity and then later upgrading capacity by adding ultra-high-capacity nodes/links. A multi-layer wireless mesh network can be deployed in other manners as well.

[0160] One variation of the multi-layer mesh architecture described above is that the ultra-high-capacity links can be designed to create specific paths based on a traffic requirement and/or some other criteria defined by the operator. To illustrate with an example, FIG. 28 shows another example of a multi-layer wireless mesh network in which some of the preexisting, non-ultra-high-capacity point-to-point links included in the example multi-layer wireless mesh network of FIG. 27 are replaced by ultra-high-capacity links (shown as triple-compound links) to provide ultra-high capacity to specific segments of the wireless mesh network. This can be done either by supplementing the hardware of the preexisting, non-ultra-high-capacity nodes at the customer location with new hardware (e.g., a new radio or other associated hardware) capable of establishing ultra-high-capacity links or by replacing the hardware of the preexisting, non-ultra-high-capacity nodes at the customer location with new hardware (e.g., a new radio or other associated hardware) capable of establishing ultra-high-capacity links.

[0161] Another variation of the multi-layer mesh architecture described above is that different layers of the wireless mesh network may be deployed at different heights, which may create physical-link separation by allowing re-use of the available frequency spectrum. For instance, in one implementation, a multi-layer wireless mesh network can have at least 2 layers of ultra-high-capacity links operating in the same frequency range, but at different heights. To illustrate with an example, a first layer of ultra-high-capacity links can be deployed at a lower height, such as by installing the required hardware at a lower height within a structure hosting the wireless mesh hardware (e.g., on a lower floor of a building), and a second layer of the ultra-high-capacity links can be deployed at a higher height, such as by installing the required hardware at a higher height of the structure hosting the wireless mesh hardware (e.g., at higher floor of the building). In this respect, the deployment of these different layers of ultra-high-capacity links at different heights may serve to increase the capacity of the multi-layer wireless mesh network.

[0162] While the foregoing example involves the deployment of multiple different layers of ultra-high-capacity links at multiple different heights, it should be understood that this example is merely provided for purposes of illustration, and that multiple layers of wireless mesh links of any type may be deployed at different heights in order to enhance the overall capacity of the multi-layer wireless mesh network, including but not limited to layers of ultra-high-capacity links, non-ultra-high-capacity point-to-point links, and/or non-ultra-high-capacity point-to-multipoint links.

[0163] Yet another variation of the multi-layer mesh architecture described above is that the point-to-multipoint links that are not ultra-high capacity (which are shown in FIGS. 27 and 28 as single-line links) may be replaced by wired links, such as a coaxial wire loop, fiber loop or some other type of wired link. To illustrate with an example, a multi-layer mesh network may include wired links that comprise the coaxial portion of the HFC (Hybrid Fiber Coax) used by the cable companies, in which case this coaxial portion of the HFC may bring mesh network connectivity to end users while the fiber portion of the HFC may bring the high-speed internet to the neighborhood. In this respect, the wireless mesh links consisting of ultra-high-capacity links (which are shown in FIGS. 27 and 28 as triple-compound links) and/or non-ultra-high-capacity point-to-point links may play the

role of the fiber equivalent portion of the HFC by bring high capacity from a fiber POP to the neighborhood.

[0164] Example embodiments of the disclosed innovations have been described above. As noted above, it should be understood that the figures are provided for the purpose of illustration and description only and that various components (e.g., modules) illustrated in the figures above can be added, removed, and/or rearranged into different configurations, or utilized as a basis for modifying and/or designing other configurations for carrying out the example operations disclosed herein. In this respect, those skilled in the art will understand that changes and modifications may be made to the embodiments described above without departing from the true scope and spirit of the present invention, which will be defined by the claims.

[0165] Further, to the extent that examples described herein involve operations performed or initiated by actors, such as humans, operators, users or other entities, this is for purposes of example and explanation only. Claims should not be construed as requiring action by such actors unless explicitly recited in claim language.

1. A mesh-based communication system for providing a mesh-based service to customer sites within a given geographic region, the mesh-based communication system comprising:

- a wireless mesh network formed by a set of wireless communication nodes, wherein the set of wireless communication nodes comprises:
 - a first tier of communication nodes that are each installed at a respective point of presence (PoP) site that provides high-capacity access to a core network;
 - a second tier of communication nodes that are each configured to extend the high-capacity access to the core network from the first tier of communication nodes to one or more other geographic locations within the given geographic region by forming one or more high-capacity pathways for routing aggregated network traffic that originated from or is destined to the core network; and
 - a third tier of wireless communication nodes that are each configured to extend the wireless mesh network from the second tier of communication nodes by forming one or more pathways for routing aggregated network traffic to and from endpoints within a particular portion of the given geographic region.

2. The mesh-based communication system of claim 1, wherein each communication node in the first tier of communication nodes is configured to (i) exchange bi-directional network traffic with the core network via a high-capacity fiber connection between the POP site and the core network, and (ii) exchange bi-directional network traffic with one or more other wireless communication nodes in another tier of communication nodes from the set of wireless communication nodes via one or more bi-directional point-to-point (ptp) or point-to-multipoint (ptmp) wireless links.

3. The mesh-based communication system of claim 2, wherein each of the one or more bi-directional ptp or ptmp wireless links comprises a millimeter-wave link.

4. The mesh-based communication system of claim 1, wherein the first tier of communication nodes comprises a subset of communication nodes, wherein each communication node in the subset of communication nodes is configured to deliver the mesh-based service to the PoP site.

5. The mesh-based communication system of claim 1, wherein each communication node in the second tier of communication nodes is (i) directly connected to a communication node in the first tier of communication nodes via a bi-directional point-to-point (ptp) or point-to-multipoint (ptmp) wireless link but not directly connected to any other communication node in the second tier of communication nodes, (ii) directly connected to a communication node in the first tier of communication nodes via a first bi-directional ptp or ptmp wireless link and also directly connected to another communication node in the second tier of communication nodes via a second bi-directional ptp or ptmp wireless link, or (iii) directly connected to two other communication nodes in the second tier of communication nodes via respective bi-directional ptp or ptmp wireless links.

6. The mesh-based communication system of claim 1, wherein each communication node in the second tier of communication nodes is configured to exchange bi-directional network traffic along the one or more high-capacity pathways either (i) with a single other communication node of the set of wireless communication nodes or (ii) with each of two other communication nodes of the set of wireless communication nodes.

7. The mesh-based communication system of claim 1, wherein the second tier of communication nodes comprises a subset of communication nodes, wherein each communication node in the subset of communication nodes is configured to directly connect downstream to one or more communication nodes in the third tier of communication nodes via one or more bi-directional point-to-point (ptp) or point-to-multipoint (ptmp) wireless links.

8. The mesh-based communication system of claim 7, wherein each of the one or more bi-directional ptp or ptmp wireless links comprises a millimeter-wave link.

9. The mesh-based communication system of claim 1, wherein the second tier of communication nodes comprises a subset of communication nodes, wherein each communication node in the subset of communication nodes is configured to deliver the mesh-based service to an infrastructure site at which the communication node is installed.

10. The mesh-based communication system of claim 1, wherein each communication node in the third tier of communication nodes is directly connected to one or more other nodes within one or more of the second and third tiers via one or more bi-directional point-to-point (ptp) or point-to-multipoint (ptmp) wireless links.

11. The mesh-based communication system of claim 10, wherein each of the one or more bi-directional ptp or ptmp wireless links comprises a millimeter-wave link.

12. The mesh-based communication system of claim 1, wherein the third tier of communication nodes comprises a subset of communication nodes, wherein each communication node in the subset of communication nodes is configured to deliver the mesh-based service to an infrastructure site at which the communication node is installed.

13. The mesh-based communication system of claim 1, wherein the set of wireless communication nodes further comprises:

- a fourth tier of wireless communication nodes that are each configured to extend the wireless mesh network to other endpoints within the given geographic region via wireless links that originate from communication nodes in one or more of the second and third tiers of communication nodes and are to route individual network

traffic to and from the communication node in the fourth tier of communication nodes.

14. The mesh-based communication system of claim 13, wherein each communication node in the fourth tier of communication nodes is (i) directly connected upstream to at least one communication node in the second tier or third tier of communication nodes via one or more bi-directional point-to-point (ptp) or point-to-multipoint (ptmp) wireless links, and (ii) configured to exchange bi-directional network traffic with the at least one communication node in the second tier or third tier of communication nodes.

15. The mesh-based communication system of claim 14, wherein each of the one or more bi-directional ptp or ptmp wireless links comprises a millimeter-wave link.

16. The mesh-based communication system of claim 1, wherein the fourth tier of communication nodes comprises a subset of communication nodes, wherein each communication node in the subset of communication nodes is configured to deliver the mesh-based service to an infrastructure site at which the communication node is installed.

17. The mesh-based communication system of claim 1, wherein each communication node of at least a subset of the set of wireless communication nodes is installed at a residence of a respective end user of the mesh-based service provided by the mesh-based communication system.

18. A mesh-based communication system for providing a mesh-based service to customer sites within a given geographic region, the mesh-based communication system comprising:

- a wireless mesh network formed by a set of wireless communication nodes, wherein the set of wireless communication nodes comprises:
 - a first tier of communication nodes that are each installed at a respective point of presence (PoP) site that provides high-capacity access to a core network;
 - a second tier of communication nodes that are each configured to extend the high-capacity access to the core network from the first tier of communication nodes to one or more other geographic locations within the given geographic region by forming one or more high-capacity pathways for routing aggregated network traffic that originated from or is destined to the core network; and
 - a third tier of wireless communication nodes that are each configured to extend the wireless mesh network from the second tier of communication nodes by forming one or more pathways for routing aggregated network traffic to and from endpoints within a particular portion of the given geographic region; and
 - a fourth tier of wireless communication nodes that are each configured to extend the wireless mesh network to other endpoints within the given geographic region via wireless links that originate from communication nodes in one or more of the second and third tiers of communication nodes and are to route individual network traffic to and from the communication node in the fourth tier of communication nodes,

wherein each communication node of at least a subset of the set of wireless communication nodes is installed at a residence of a respective end user of the mesh-based service provided by the mesh-based communication system.

19. The mesh-based communication system of claim **18**, wherein each communication node in the second tier of communication nodes is (i) directly connected to a communication node in the first tier of communication nodes via a bi-directional point-to-point (ptp) or point-to-multipoint (ptmp) wireless link but not directly connected to any other communication node in the second tier of communication nodes, (ii) directly connected to a communication node in the first tier of communication nodes via a first bi-directional ptp or ptmp wireless link and also directly connected to another communication node in the second tier of communication nodes via a second bi-directional ptp or ptmp wireless link, or (iii) directly connected to two other communication nodes in the second tier of communication nodes via respective bi-directional ptp or ptmp wireless links.

20. The mesh-based communication system of claim **18**, wherein each communication node in the third tier of communication nodes is directly connected to one or more other nodes within one or more of the second and third tiers via one or more bi-directional point-to-point (ptp) or point-to-multipoint (ptmp) wireless links.

* * * * *